

ASM

PROPRIETARY PERFORMANCE SYSTEMS

ENGINEERING CONCEPT MANUAL | REVISION A



PURPOSE

A development blueprint for twenty-two ASM-only braking, wheel, aerodynamic, suspension, thermal, controls, powertrain, structural, telemetry, and hardware systems. Includes concept schematics, design variables, material/process candidates, prototype pathways, failure-mode controls, and verification gates.

CONCEPT LEVEL - ENGINEERING VALIDATION REQUIRED

ASM-EPM-001 | 2026-07-31 | REV A

DOCUMENT STATUS AND LIMITATIONS

This manual is a concept-development document, not a set of certified fabrication drawings. It defines architectures, parameter sets, materials/process candidates, schematic interfaces, test pathways, and release evidence. It deliberately does not supply universal dimensions, torque values, thread sizes, laminate schedules, weld procedures, heat treatments, or safety limits because those values must be established for a specific vehicle, ruleset, load history, supplier process, and legal market.

INTENDED USE

- Establish ASM product architecture and visual/engineering DNA.
- Create requirements and interface-control documents.
- Commission professional design, simulation, prototype, and test work.
- Build a defensible design history and intellectual-property package.
- Prevent concept art from being mistaken for production engineering.

REQUIRED APPROVALS

A qualified mechanical/automotive engineer must approve structural and thermal design. Brake friction and hydraulic parts require specialist supplier and dynamometer validation. Composite structures require approved laminate/process design. Electronics require functional-safety and cybersecurity scoping. Competition and road use require the applicable rules, homologation, and legal review.

ASM COMMON DESIGN DNA

- Halo Spine: visible structural arch/rib or internal load path.
- Aero Veins: directional flow/heat/stiffness channels.
- Trinity Geometry: three-point visual and service interfaces where function permits.
- Component Passport: identity, revision, material lot, life, setup, inspection, and event history.
- Red health cue: reserved for lock, thermal, adjustment, or safety state - never decoration alone.

CONFIGURATION RULE

A product name is not a configuration. Every usable configuration must identify vehicle application, revision, option set, material/process route, software/calibration, approved companion parts, installation procedure, and operating limits.

CONTROLLED DEVELOPMENT LANGUAGE

Term	Meaning
Concept	Architecture is described; loads, dimensions, and processes remain provisional.
Development mule	Prototype intended to learn; not representative of final durability or production quality.
DV	Design validation: proves the design can meet requirements using controlled test articles.
PV	Production validation: proves the manufacturing process repeatedly produces conforming parts.
Released	Approved configuration with signed evidence, manufacturing controls, installation/service documents, and life limits.

System Index and Product Architecture

FRONT MATTER

Code	System	Division	Primary differentiator
HB-01	HaloBridge Monobloc Brake Caliper	Stopping Dynamics	Removable HaloBridge arch, mirrored Aero Veins, Trinity service locks, and a visible load/temperature health strip.
TZ-02	Tri-Zone Brake Pad Family	Stopping Dynamics	Three functional surface zones, curved HaloCut channels, coded thermal backing plates, and a common ASM wear/temperature language.
HR-03	HaloRotor Floating Brake Disc	Stopping Dynamics	Directional HaloSweep vanes, Trinity float groups, scalloped heat-map marks, and a sensor-ready lightweight hat.
HF-04	HaloFlow Brake-Duct System	Stopping Dynamics	Dual-path Coanda splitter, numbered airflow cartridges, debris-shedding inlet, and sealed steering-motion interfaces.
HL-05	HaloLock Center-Lock System	Wheel Dynamics	Three-lobe Trinity nut, mirrored side identification, captive secondary lock, TorqueSight witness collar, and digital cycle history.
HW-06	HaloRail Center-Lock Forged Wheel	Wheel Dynamics	Structural halo ring joining selected spokes, asymmetric Aero Vein spokes, replaceable rim-edge protector, and aerodynamic center cartridges.
AS-07	AeroSkin Modular Body Panels	Aero Dynamics	Internal Halo Spine ribs, curved Aero Vein extraction slots, modular vent cartridges, flush SaintKey fasteners, and sacrificial strike rails.
HA-08	HaloArc Instrumented Racing Wing	Aero Dynamics	Subtle central HaloArc planform, swan-neck carbon/titanium pylons, indexed HaloClick adjusters, replaceable vortex tips, and strain-measured load path.
VB-09	VectorBar Dual-Rate Anti-Roll System	Cornering Dynamics	Coaxial torsion architecture, HaloIndex blade adjusters, replaceable stiffness cartridges, and optional torsion-strain telemetry.
CB-10	CornerBlade Aerodynamic Control Arms	Cornering Dynamics	Airfoil arm sections, replaceable SaintLink bearing cartridges, indexed camber inserts, internal protected routing, and optional load sensing.
SL-11	SaintLink Heave/Roll Suspension Platform	Cornering Dynamics	Central Halo heave element, separated roll path, indexed rockers, modular springs/dampers, and position/load sensing.
HU-12	HaloCore Upright and Hub Cassette	Wheel Dynamics	Topology-shaped HaloCore, keyed bearing cassette, integrated duct passages, protected sensor spine, and inspection windows.
VR-13	VortexRail Splitter, Floor, and Diffuser	Aero Dynamics	VortexRail edge geometry, load-sensing splitter stays, replaceable strake cartridges, sacrificial skid rails, and controlled-release outer blades.
TS-14	Thermal Spine Cooling Architecture	Thermal Dynamics	Central thermal backbone, modular cooling cartridges, Aero Vein exits, pressure/temperature sensing, and Sentinel setup recommendations.
TF-15	TorqueFrame Tunable Chassis Bracing	Structural Dynamics	Halo-shaped node plates, replaceable stiffness links, strain-marked load paths, and serialized setup modules.
VP-16	VectorPedal Sensorized Pedal Box	Driver Dynamics	Trinity pedal faces, indexed ratio inserts, captured balance-bar settings, modular heel plate, and dual-sensor channels.
HS-17	HaloBox Sequential Shifter	Driver Dynamics	Halo lever arch, adjustable force cartridge, indexed throw positions, load-cell handle, and protected reverse/neutral logic.
AI-18	AeroPhase Tunable Intake	Powertrain Dynamics	Halo plenum rib, numbered velocity-stack cartridges, equal-path Aero Veins, and integrated pressure/temperature instrumentation.
SF-19	SaintFlow Titanium Exhaust	Powertrain Dynamics	Equal-path collectors, Halo expansion joints, modular silencer/catalyst sections, three-element outlet motif, and integrated temperature/pressure ports.
SN-20	Sentinel Telemetry and Component Passport	Sentinel Systems	Component passports, Halo health score, configuration checksum, synchronized corner data, and service-gate alerts.
SK-21	SaintKey Controlled Hardware Family	ASM Platform	Trinity tool interface, captive elements, red/white lock witnesses, keyed orientation, and serialized critical hardware kits.
CP-22	Integrated ASM Cornering Package	ASM Platform	One setup passport linking HaloBridge, HaloFlow, HaloLock, HaloRail, VectorBar, CornerBlade, SaintLink, VortexRail, and HaloArc.

ASM Product Development Lifecycle

FRONT MATTER

THE V-MODEL DISCIPLINE

Gate	Core question	Minimum outputs
0 - Opportunity	Is the customer/vehicle problem real and measurable?	Use case; baseline data; business case; rules/market scan.
1 - Requirements	What must the system do and never do?	Numbered requirements; interfaces; hazards; acceptance methods.
2 - Architecture	Which concept best satisfies the requirements?	Trade matrix; selected concept; preliminary models; patent/FTO screen.
3 - Detail design	Can it be manufactured and inspected?	Released CAD/drawings; tolerance analysis; DFMEA; supplier/process plan.
4 - DV	Does representative hardware meet design requirements?	Approved procedures; test data; teardown; closed anomalies.
5 - PV	Can the intended process make it repeatedly?	Capability data; first article; control plan; approved work instructions.
6 - Vehicle release	Is the integrated vehicle safe, compliant, and supportable?	Integration report; track/endurance evidence; rules approval; service pack.
7 - Field learning	Does evidence support continued use?	Incident/warranty review; life updates; controlled revisions.

NO-SKIP RULE

REQUIREMENT QUALITY TEST

Each requirement should be necessary, singular, measurable, achievable, traceable, and unambiguous. Avoid words such as “strong,” “light,” “high temperature,” or “race proven” without a numeric condition and verification method.

BASELINE-FIRST TESTING

Instrument the unmodified vehicle first. Preserve baseline setup, tires, driver, environment, and test sequence. Compare on repeatable metrics and report uncertainty. A faster isolated lap is not proof of durability, stability, or customer value.

CHANGE CONTROL

Changes to geometry, material, finish, supplier, heat treatment, software, tooling, lubricant, torque procedure, or inspection method can invalidate evidence. Classify changes and define what analysis or retest is required.

INDEPENDENT REVIEW

Safety-critical releases require a reviewer who did not create the primary design. The reviewer must have access to assumptions, raw results, anomalies, and configuration records.

Common Engineering Calculation Framework

FRONT MATTER

Domain	First-order relationships	What the model must ultimately include
Braking	Energy per event; axle energy share; hydraulic clamp; effective radius; friction map.	Dynamic load transfer, ABS/control, pad/rotor thermal network, cooling, knock-back, tolerances.
Structures	Stress, strain, stiffness, buckling, bearing, joint preload, fatigue usage.	Real joint/contact stiffness, nonlinearities, measured spectra, manufacturing defects, temperature.
Aerodynamics	$q = 0.5 \rho V^2$; $F = q S C$; pressure integration; balance.	Ride/pitch/yaw/steer map, rotating wheels, cooling flow, moving ground, structural deflection.
Suspension	Motion ratio; wheel rate; roll/heave modes; compliance.	Full kinematics, joint friction, tire data, aero load, bump stops, damper temperature.
Thermal/fluids	$Q = m c \Delta T$; $\dot{Q} = \dot{m} c \Delta T$; ΔP ; convection.	Transient networks, radiation, boiling/cavitation, heat soak, fouling, pump/fan maps.
Rotating systems	Torque path, inertia, balance, bearing life, overspeed.	Joint friction, temperature, curb events, fatigue spectrum, NDT limits.
Data/software	Sampling, filtering, uncertainty, plausibility, state logic.	Clock synchronization, fault injection, cybersecurity, change management, raw-data retention.

MODEL CORRELATION

Identify quantities measurable in both simulation and test: strain, displacement, pressure, temperature, load, flow, or frequency. Update uncertain boundary conditions rather than forcing a cosmetic match.

UNCERTAINTY

Report sensor accuracy, calibration status, repeatability, environmental variation, manufacturing variation, and model sensitivity. Define guard bands when pass/fail depends on uncertain measurement.

MARGIN

Margin is requirement- and failure-mode-specific. Do not apply one generic factor to pressure, fatigue, thermal, composite, joint, and aero problems. Record the source and rationale for every limit.

Reference Framework and Compliance Mapping

FRONT MATTER

Reference family	Why it matters to ASM	Use in this manual
FIA 2026 technical regulations and Appendix J	Competition eligibility, safety construction, component restrictions, and event-specific requirements.	Rules review must occur at requirements, concept freeze, and final configuration.
NHTSA FMVSS No. 135 test procedure and interpretations	Road-vehicle light-vehicle braking requirements in the United States when applicable.	Defines a legal/compliance path separate from race-only performance testing.
SAE J2522 dynamometer practice	Disc-brake effectiveness evaluation on an inertia dynamometer.	Useful framework for friction-system mapping; exact project procedure must be approved.
SAE J2530 aftermarket wheel performance	Aftermarket wheel test framework including fatigue/impact concepts.	Input to wheel DV planning; application and current revision must be confirmed.
UNECE Regulation No. 90	Replacement brake lining/disc/drum requirements in relevant markets.	Commercial road-product pathway; not replaced by race-track tests.
ISO 26262:2018	Functional safety for road-vehicle electrical/electronic systems.	Scope Sentinel and sensor-dependent functions; mechanical safety remains independent.
ISO/SAE 21434:2021	Road-vehicle cybersecurity engineering.	Threat analysis, secure updates, access, and lifecycle controls for connected data.
ASME Y14.5	Dimensioning and tolerancing language.	Basis for unambiguous drawings and datum/tolerance communication.

REFERENCE CONTROL NOTE

Standards and rules change. The design record must identify the exact edition/date, jurisdiction, vehicle classification, sanctioning body, and approved interpretation used. A standards citation is not proof of compliance; compliance requires a documented applicability assessment and evidence against each applicable clause.

Facilities, Partners, and Core Team

FRONT MATTER

CAPABILITY ROADMAP

CORE RESPONSIBILITIES

Capability	Minimum purpose	Likely make/buy strategy	Role	Accountability
3D scan, CAD, PLM, drawing control	Master geometry, interfaces, revisions, and released data.	Own core digital thread; use spec metrology as needed.	Chief engineer	Requirements, architecture, margins, integration, and release recommendations.
FEA/CFD/1D/multibody tools	Loads, stiffness, thermal, flow, aero, and kinematics.	Own requirements/models; peer-reviewed specialists.	Design engineer	CAD, drawings, calculations, tolerance analysis, and design changes.
Brake inertia dynamometer	Friction, thermal, fade, recovery, wear, and compatibility.	Partner with established brake test laboratory.	Test/validation lead	DV/PV strategy, procedures, calibration, anomalies, and signed reports.
Structural rigs and servo-hydraulic fatigue	Proof, fatigue, joints, arms, uprights, wings, wheels.	Partner initially; build modular fix in-house.	Manufacturing/quality engineer	Supplier process, control plan, FAI, capability, nonconformance, and general compliance.
Composite shop and NDT	Controlled laminate/tooling/cure/inspection.	Qualified supplier until volume justifies controlled facility.	Vehicle dynamics/aero/thermal specialists	Domain models, test correlation, and operating envelopes.
Machine/forge/cast/heat-treat/NDT supply chain	Production-intent metallic components.	Approved suppliers with process traceability.	Functional-safety/cyber lead	Safety concept and cyber lifecycle for electronic/connected functions.
K&C/corner rig, damper dyno, flow bench	Suspension/steering compliance, dampers, ducts/intake.	Blend internal bench tools with external labs.	Independent reviewer	Challenges assumptions and approves safety-critical evidence.
Instrumented development vehicle and closed test venue	System correlation, driver feedback, endurance, safe failure assessment.	ASM-owned mule; professional track/test operations.		

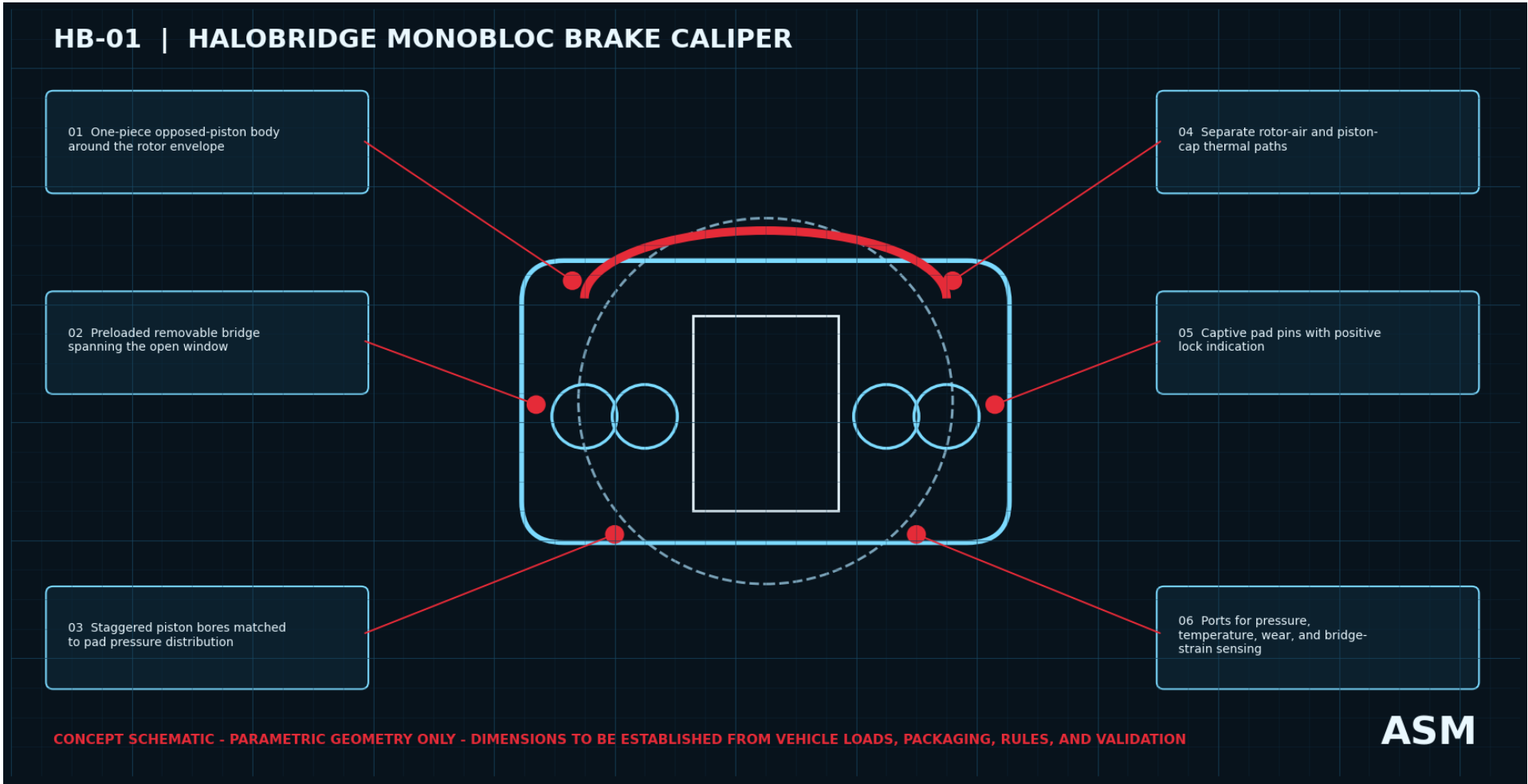
SUPPLIER RULE

HaloBridge Monobloc Brake Caliper

HB-01 / SHEET 1 OF 4

STOPPING DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Create a rigid, serviceable, sensor-ready fixed caliper whose external preloaded bridge and directional cooling architecture are unmistakably ASM.	Removable HaloBridge arch, mirrored Aero Veins, Trinity service locks, and a visible load/temperature health strip.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloBridge Monobloc Brake Caliper - Architecture and Design Basis

HB-01 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- One-piece opposed-piston body around the rotor envelope
- Preloaded removable bridge spanning the open window
- Staggered piston bores matched to pad pressure distribution
- Separate rotor-air and piston-cap thermal paths
- Captive pad pins with positive lock indication
- Ports for pressure, temperature, wear, and bridge-strain sensing

PRIMARY DESIGN VARIABLES

- Effective piston area and front/rear hydraulic balance
- Bridge preload, body opening deflection, and seal-groove stability
- Rotor/pad envelope, wheel-barrel clearance, and steering sweep
- Piston diameter progression and knock-back allowance
- Fluid passage pressure drop, bleed orientation, and residual air volume
- Thermal path from pad to piston, seal, fluid, bridge, and wheel

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Upright/caliper bracket
- Rotor annulus and axial float
- Brake hose hard-point and steering travel
- Wheel barrel and balance-weight envelope
- Pad family and retention hardware
- CAN/data logger through protected sensor harness

CALCULATION FRAMEWORK

- Clamp force: $F_c = P \cdot h \cdot A_{total}$
- Brake torque: $T_b = 2 \cdot \mu \cdot F_{side} \cdot R_{eff}$
- Bridge stiffness target from $k = F/\delta$
- Thermal energy per stop: $E = 0.5 \cdot m \cdot (v1^2 - v2^2)$ times axle share

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloBridge Monobloc Brake Caliper - Materials, Manufacturing, and Quality

HB-01 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Body	Forged 7075-T73 or qualified high-strength aluminum	High stiffness-to-mass; corrosion and fatigue validation required
Bridge	Ti-6Al-4V or precipitation-hardened steel	High stiffness and preload retention
Pistons	Titanium, stainless, or aluminum with ceramic caps	Select by conductivity, galling, mass, and seal compatibility
Seals	Brake-fluid-compatible high-temperature elastomer	Supplier-qualified only

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Near-net forging or billet development mule	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	5-axis machining with datum-controlled bore operations	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Stress relief, surface conversion/anodize where compatible	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Honed bores, controlled seal-groove finish, clean-room assembly	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Proof pressure, leakage, piston retraction, and dimensional CMM report	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloBridge Monobloc Brake Caliper - Verification, Failure Modes, and Release

HB-01 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	FEA: pressure, torque reaction, bridge preload, fatigue, and thermal distortion	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Hydraulic proof and burst testing with controlled fluid and temperature	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Caliper dynamometer thermal cycle with instrumented bridge deflection	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Leakage, seal roll-back, piston corrosion, pad change, and misuse tests	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Vehicle brake balance, knock-back, ABS interaction, and endurance validation	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Body cracking or yielding	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bridge fastener relaxation	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Seal overheating or extrusion	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Uneven pad pressure/taper wear	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Piston seizure or corrosion	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Air trapping and incomplete bleeding	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ No permanent deformation at proof load
- ☐ Defined maximum bridge opening under peak line pressure
- ☐ Stable release clearance through thermal cycle
- ☐ Repeatable pad replacement without loose hardware
- ☐ Traceable life limit and inspection criteria

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

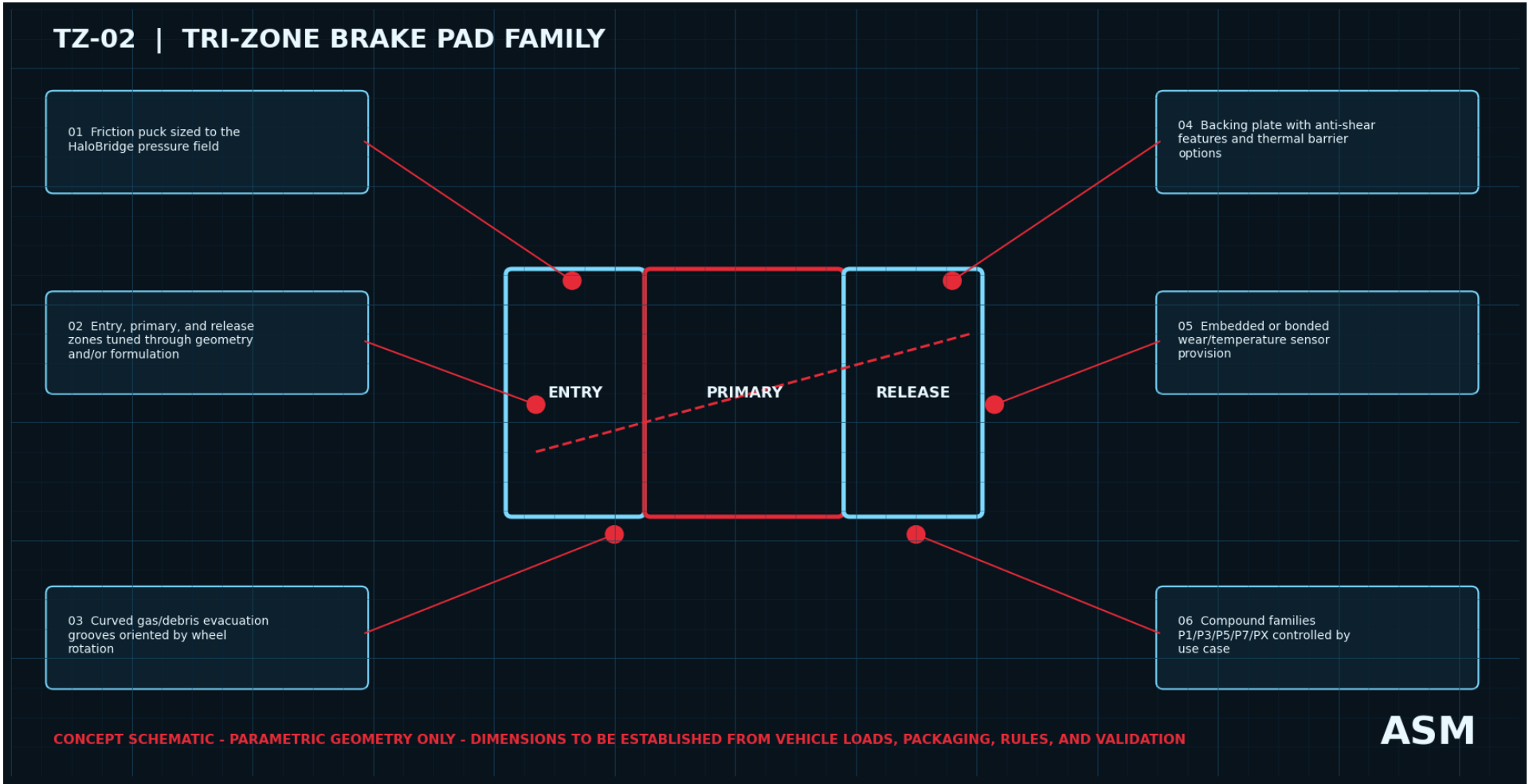
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

Tri-Zone Brake Pad Family

TZ-02 / SHEET 1 OF 4

STOPPING DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Develop a family of ASM friction materials and backing systems that provide a tunable pedal signature without sacrificing fade resistance or release consistency.	Three functional surface zones, curved HaloCut channels, coded thermal backing plates, and a common ASM wear/temperature language.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

Tri-Zone Brake Pad Family - Architecture and Design Basis

TZ-02 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Friction puck sized to the HaloBridge pressure field
- Entry, primary, and release zones tuned through geometry and/or formulation
- Curved gas/debris evacuation grooves oriented by wheel rotation
- Backing plate with anti-shear features and thermal barrier options
- Embedded or bonded wear/temperature sensor provision
- Compound families P1/P3/P5/P7/PX controlled by use case

PRIMARY DESIGN VARIABLES

- Nominal and hot coefficient of friction versus pressure, speed, and temperature
- Compressibility, release hysteresis, and pedal travel contribution
- Specific wear rate and rotor aggressiveness
- Backing-plate flatness and shear strength
- Thermal conductivity into pistons and fluid
- Noise propensity and street legality where applicable

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Caliper piston footprint
- Rotor metallurgy and surface finish
- ABS/brake-by-wire control behavior
- Cooling airflow and expected duty cycle
- Street noise/dust requirements versus race-only performance

CALCULATION FRAMEWORK

- Friction work: $W = \int T_b \, d\theta$
- Average heat flux: $q = E / (A_{\text{swept}} t_{\text{stop}})$
- Pad life estimate from wear rate times pressure-speed-temperature duty

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

Tri-Zone Brake Pad Family - Materials, Manufacturing, and Quality

TZ-02 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Friction material	Supplier-developed resin/metal/ceramic composite	Requires dyno formulation development and environmental controls
Backing plate	High-strength low-alloy steel or stainless	Flatness, corrosion, and mechanical retention critical
Barrier	Titanium shim, mica/ceramic laminate, or coated steel	Tune conductivity and compressibility
Sensor	High-temperature loop or thermocouple	Protected, replaceable, and fail-detectable

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Develop with an established friction-material formulator	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Tooling for puck molding, curing, grinding, scorching, and slotting	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Mechanical retention geometry plus qualified adhesive where used	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Batch coupon testing and material genealogy	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Final thickness/parallelism inspection and axle-set pairing	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

Tri-Zone Brake Pad Family - Verification, Failure Modes, and Release

TZ-02 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Full-scale inertia dynamometer mapping over pressure/speed/temperature	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Fade, recovery, green effectiveness, wear, and judder screening	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Shear and bond tests after thermal aging and fluid contamination	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Rotor compatibility, crack initiation, transfer-layer stability	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Vehicle subjective modulation plus objective deceleration/pressure maps	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Friction fade or unpredictable recovery	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Backing-plate separation	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Excessive compressibility	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Taper or radial wear	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Thermal cracking/chunking	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Rotor scoring or unstable transfer layer	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Published friction envelope by compound
- ☐ Controlled axle-set thickness and compressibility
- ☐ No loss of retention after worst-case thermal cycle
- ☐ Defined discard thickness and sensor trigger
- ☐ Batch traceability to raw material lot

RELEASE HOLD POINTS

1. Independent calculation/model review.
2. Drawing and tolerance review.
3. Supplier process-readiness review.
4. Design validation report approved.
5. Production validation on representative parts.
6. Vehicle integration and rules/compliance review.
7. Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

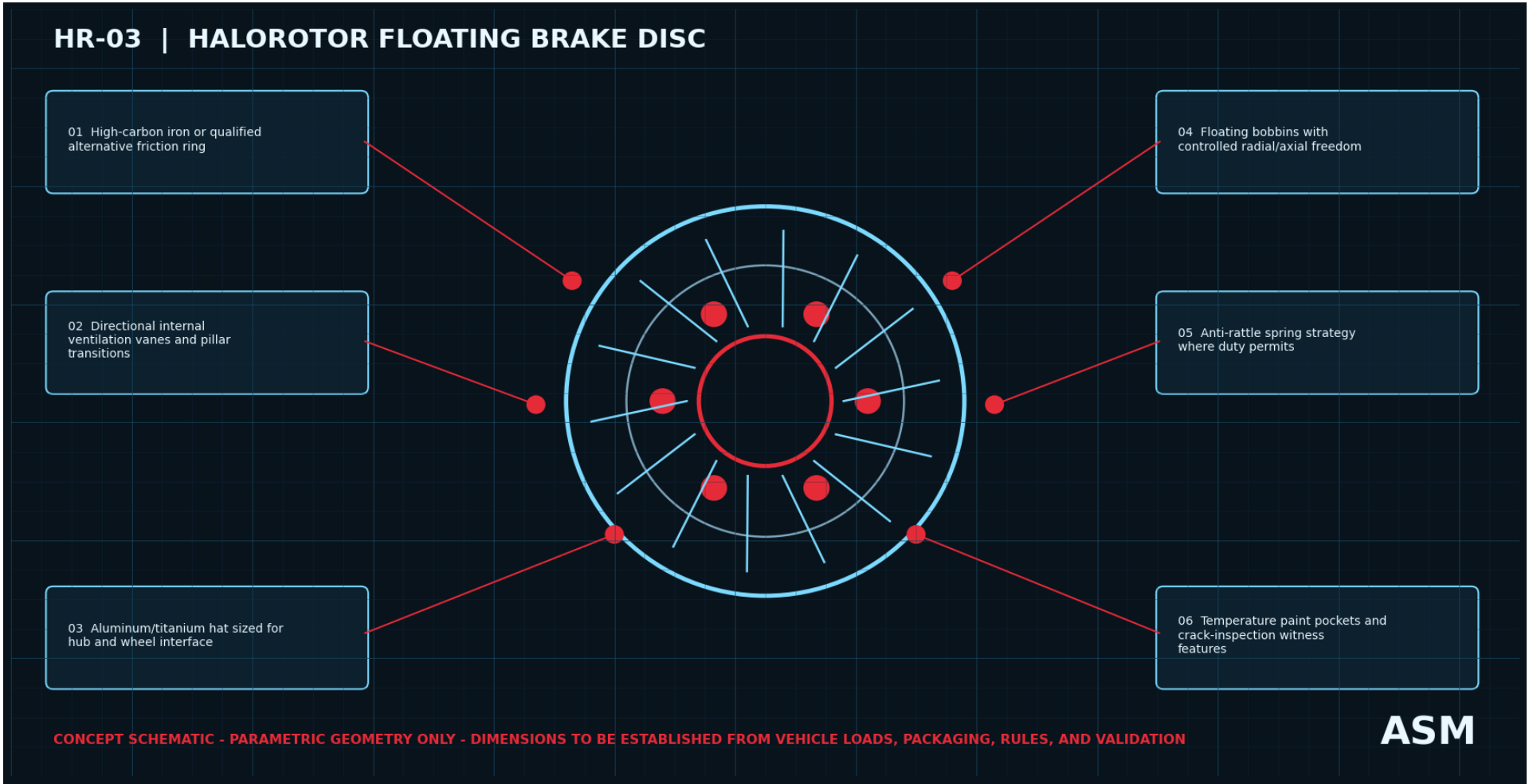
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

HaloRotor Floating Brake Disc

HR-03 / SHEET 1 OF 4

STOPPING DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Create a thermally stable floating rotor with ASM-specific vane, hat, and bobbin architecture matched to HaloBridge and HaloFlow.	Directional HaloSweep vanes, Trinity float groups, scalloped heat-map marks, and a sensor-ready lightweight hat.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloRotor Floating Brake Disc - Architecture and Design Basis

HR-03 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- High-carbon iron or qualified alternative friction ring
- Directional internal ventilation vanes and pillar transitions
- Aluminum/titanium hat sized for hub and wheel interface
- Floating bobbins with controlled radial/axial freedom
- Anti-rattle spring strategy where duty permits
- Temperature paint pockets and crack-inspection witness features

PRIMARY DESIGN VARIABLES

- Outer diameter, thickness, annulus, and effective radius
- Vane count, passage hydraulic diameter, and pumping direction
- Thermal coning, hot spotting, and hoop stress
- Hat/bobbin load transfer and differential expansion
- Unsprung inertia and polar moment
- Pad overhang, edge margin, and wear allowance

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Hub pilot and drive face
- Caliper radial/axial envelope
- Wheel spoke and barrel temperature exposure
- Duct nozzle targeting rotor eye
- ABS tone ring if integrated

CALCULATION FRAMEWORK

- Brake torque: $T = 2 \mu P A R_{eff}$
- Rotor energy rise: $\Delta T \approx E / (m c_p)$
- Thermal stress screening: σ proportional to $E_{mod} \alpha \Delta T$
- Air pumping trend from pressure rise and vane flow

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloRotor Floating Brake Disc - Materials, Manufacturing, and Quality

HR-03 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Ring	High-carbon gray iron, motorsport-qualified	Friction stability, heat capacity, crack resistance
Hat	7075-T6/T73 aluminum or titanium	Temperature and fatigue dependent
Bobbins	Stainless or titanium with hardened wear faces	Float control and galling resistance
Springs	High-temperature stainless	Retention after cycling

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Cast/heat-treated ring from qualified foundry	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	CNC machine faces, OD/ID, slots, and bobbin windows	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Balance each ring/assembly and verify runout	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Machine/anodize or coat hat with masked critical joints	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Assemble using controlled float stack and serialized hardware	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloRotor Floating Brake Disc - Verification, Failure Modes, and Release

HR-03 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Thermal FEA with convection/radiation and repeated stops	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Spin proof and overspeed balance test	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Dynamometer crack, coning, DTV, and judder program	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Bobbins: wear, seizure, impact, and contamination	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Vehicle airflow correlation using pressure/temperature instrumentation	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Thermal crack propagation	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Excessive coning or DTV	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bobbin seizure or wear	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Hat fatigue at drive lugs	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Balance loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Vane blockage or incorrect handed installation	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Defined crack acceptance/discard map
- ☐ Controlled cold and hot runout
- ☐ No bobbin bind across full temperature range
- ☐ Minimum ring thickness and life tracking
- ☐ Clear left/right identification

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

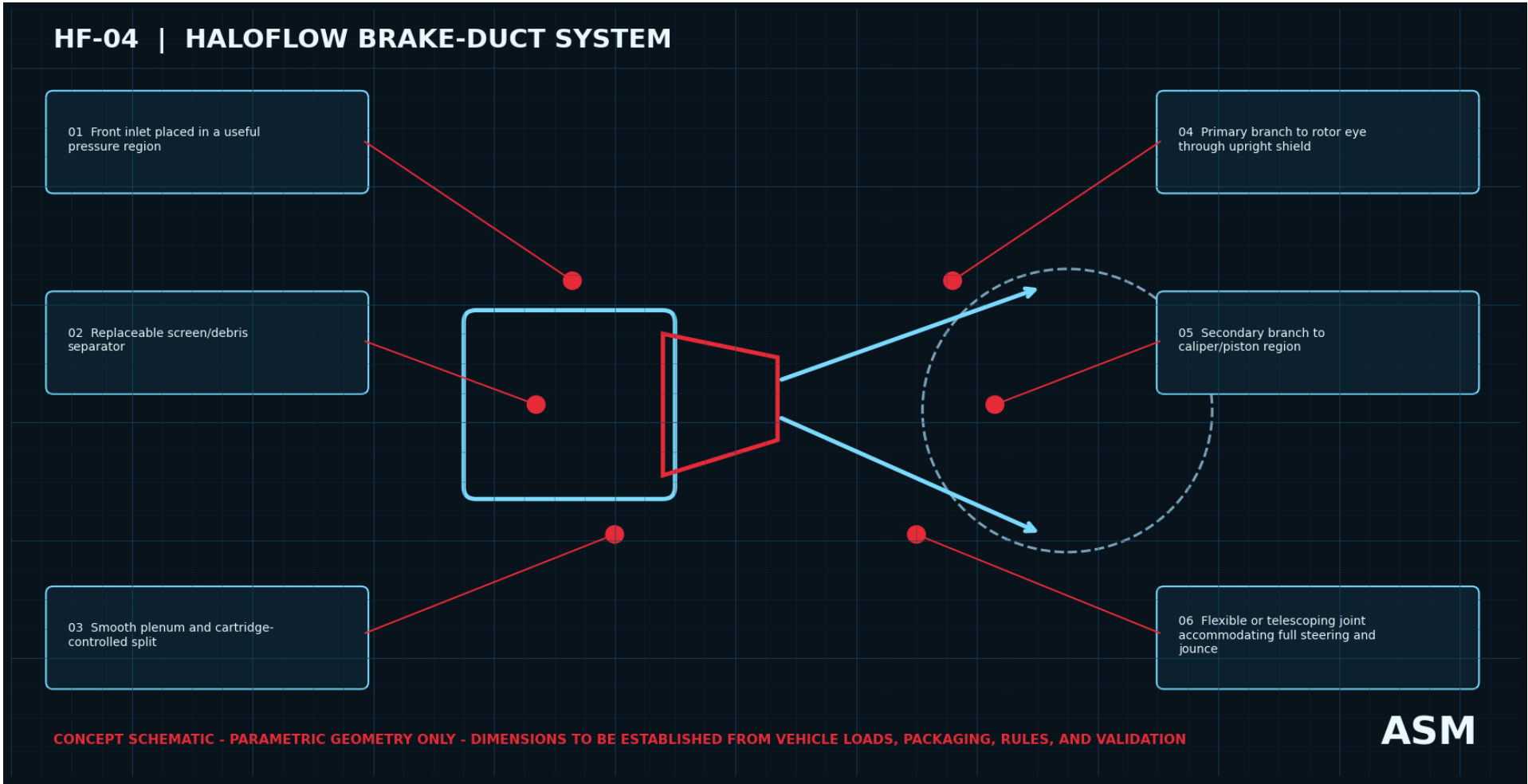
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

HaloFlow Brake-Duct System

HF-04 / SHEET 1 OF 4

STOPPING DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Deliver tunable, low-loss brake cooling to the rotor eye and caliper without fragile generic hoses or uncontrolled leakage.	Dual-path Coanda splitter, numbered airflow cartridges, debris-shedding inlet, and sealed steering-motion interfaces.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloFlow Brake-Duct System - Architecture and Design Basis

HF-04 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Front inlet placed in a useful pressure region
- Replaceable screen/debris separator
- Smooth plenum and cartridge-controlled split
- Primary branch to rotor eye through upright shield
- Secondary branch to caliper/piston region
- Flexible or telescoping joint accommodating full steering and jounce

PRIMARY DESIGN VARIABLES

- Required mass flow for rotor and caliper temperature targets
- Inlet pressure coefficient and duct pressure loss
- Minimum bend radius and area transitions
- Wheel/tyre rub envelope across all travel
- Water/debris behavior and screen blockage
- Heat resistance and impact release strategy

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Body inlet and aero package
- Steering knuckle/upright
- Control arms and tie rod
- Rotor vane inlet geometry
- Wheel barrel and tyre envelope

CALCULATION FRAMEWORK

- Mass flow: $\dot{m} = \rho A V$
- Pressure loss: $\Delta P = K \rho V^2/2$
- Cooling: $\dot{Q} = \dot{m} c_p (T_{out} - T_{in})$
- Convective trend: $\dot{Q} = h A (T_{surface} - T_{air})$

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloFlow Brake-Duct System - Materials, Manufacturing, and Quality

HF-04 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Main ducts	Pre-preg carbon or high-temperature thermoplastic	Thin, stiff, replaceable sections
Flexible joint	Silicone-coated glass or qualified high-temp hose	Fatigue and abrasion validation
Nozzle/shield	Aluminum, stainless, or composite	Temperature/impact dependent
Cartridge	Machined polymer/aluminum or printed high-temp polymer	Fast trackside changes

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	CFD concept study then SLA/SLS prototype	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Pressure-tap bench flow test for every cartridge	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Composite tooling and bonded flange development	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Vehicle scan and kinematic sweep fixture	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Witness coating to detect rubbing and heat exposure	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloFlow Brake-Duct System - Verification, Failure Modes, and Release

HF-04 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	CFD plus bench flow correlation at several yaw/steering states	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Wheel-house pressure mapping on vehicle	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Thermal track test with blanked/low/medium/high cartridges	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Debris, water, curb strike, and hose fatigue tests	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Failure-state check: detached duct must not contact tyre or steering	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Duct collapse or blockage	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Tyre/wheel contact	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Nozzle detachment	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Flow imbalance left/right	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Water ingestion shock	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Excess cooling causing drag or pad under-temperature	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Cartridge flow map and visual ID
- ☐ No contact through full kinematic envelope
- ☐ Controlled breakaway without loose dangerous debris
- ☐ Repeatable temperature reduction with measured pressure penalty

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

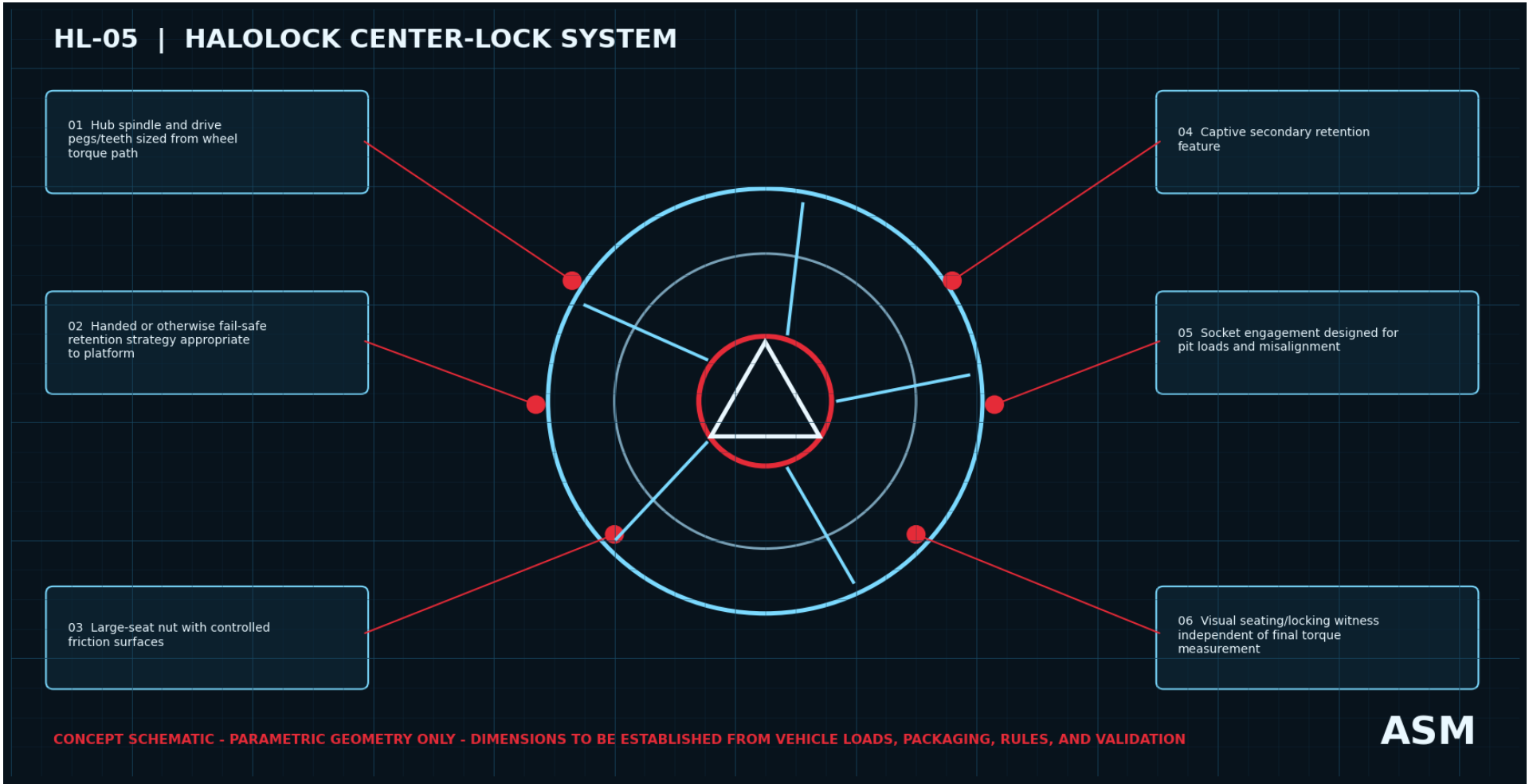
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

HaloLock Center-Lock System

HL-05 / SHEET 1 OF 4

WHEEL DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Create a fast, inspectable, life-tracked single-nut wheel retention system with redundant human-factors controls.	Three-lobe Trinity nut, mirrored side identification, captive secondary lock, TorqueSight witness collar, and digital cycle history.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloLock Center-Lock System - Architecture and Design Basis

HL-05 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Hub spindle and drive pegs/teeth sized from wheel torque path
- Handed or otherwise fail-safe retention strategy appropriate to platform
- Large-seat nut with controlled friction surfaces
- Captive secondary retention feature
- Socket engagement designed for pit loads and misalignment
- Visual seating/locking witness independent of final torque measurement

PRIMARY DESIGN VARIABLES

- Clamp load versus applied torque and friction scatter
- Thread form, pitch, engagement, and root fatigue
- Nut seat pressure and wheel-center embedment
- Drive feature shear/bearing stress
- Thermal growth across hub, wheel, and nut
- Socket reaction, pit-gun speed, and cross-thread prevention

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Wheel center and drive pockets
- Bearing cartridge and upright
- Brake rotor hat/tone ring
- Pit socket and torque tool
- Component Passport and service workflow

CALCULATION FRAMEWORK

- Approximate clamp: $F = T / (K d)$, determine K experimentally
- Drive shear: $\tau = \text{Torque} / (n r A)$
- Thread bearing and root fatigue by validated joint model
- Thermal differential: $\Delta T_{\text{al}} = \alpha L \Delta T$

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloLock Center-Lock System - Materials, Manufacturing, and Quality

HL-05 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Spindle	Ultra-high-strength alloy steel, heat treated and shot peened	Fatigue, notch, plating, and hydrogen controls
Nut	Titanium or qualified high-strength alloy	Mass, galling, temperature, and thread life
Drive features	Hardened steel or titanium inserts	Replaceability desirable
Safety collar	Heat-resistant alloy with positive detent	Must remain captive

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Thread/grind critical surfaces after heat treatment as required	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	100% magnetic-particle or penetrant inspection by material	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Gauge thread and seat geometry; measure prevailing features	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Instrumented torque-to-clamp development with actual coatings/lubricant	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Proof/load-cycle and overspeed testing with wheel/hub assembly	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloLock Center-Lock System - Verification, Failure Modes, and Release

HL-05 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Joint FEA including friction/contact and bending	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Torque-clamp scatter across coatings, reuse cycles, contamination, and heat	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Rotational fatigue, curb impact, overspeed, and abuse	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Pit-stop trials for seating, cross-thread, partial lock, and error recognition	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	NDT interval validation and retirement criteria	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Loss of preload	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Thread galling/cross-thread	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Drive-feature yielding	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Nut loosening	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Spindle fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Incorrect side installation	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Two independent confirmation cues
- ☐ Defined lubrication and torque procedure
- ☐ Cycle/temperature life limits
- ☐ No single loose secondary component
- ☐ Keyed or unmistakable left/right identification

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

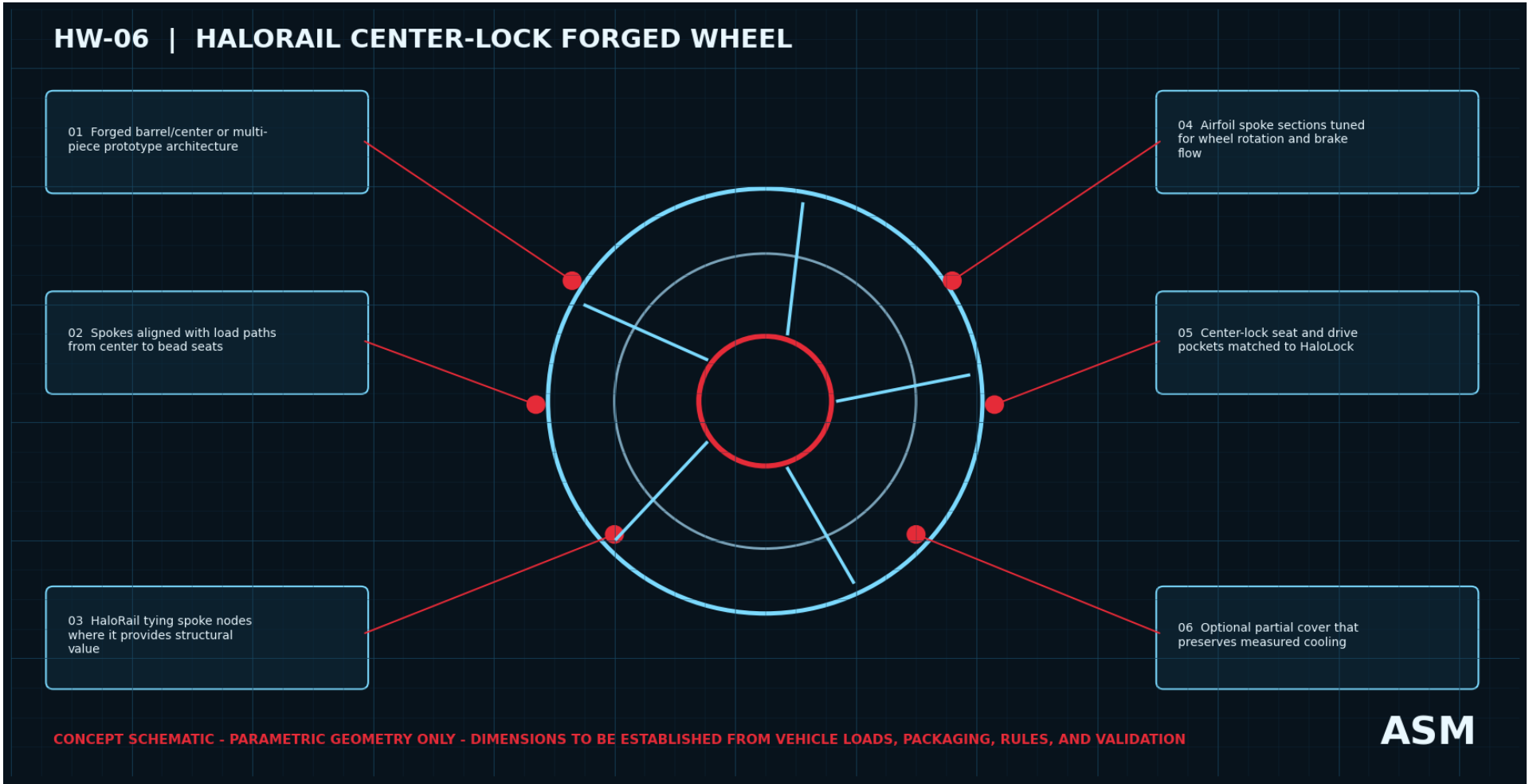
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

HaloRail Center-Lock Forged Wheel

HW-06 / SHEET 1 OF 4

WHEEL DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Develop a lightweight forged wheel that actively manages brake airflow and carries an identifiable continuous HaloRail load path.	Structural halo ring joining selected spokes, asymmetric Aero Vein spokes, replaceable rim-edge protector, and aerodynamic center cartridges.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloRail Center-Lock Forged Wheel - Architecture and Design Basis

HW-06 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Forged barrel/center or multi-piece prototype architecture
- Spokes aligned with load paths from center to bead seats
- HaloRail tying spoke nodes where it provides structural value
- Airfoil spoke sections tuned for wheel rotation and brake flow
- Center-lock seat and drive pockets matched to HaloLock
- Optional partial cover that preserves measured cooling

PRIMARY DESIGN VARIABLES

- Static radial, cornering, torque, and impact loads
- Fatigue spectrum from target vehicle and track curbs
- Bead-seat, flange, and drop-center geometry
- Center bore/seat concentricity and drive-pocket load sharing
- Mass, polar inertia, and balance correction allowance
- Brake, caliper, valve, tyre, and suspension clearances

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- HaloLock nut/seat/drive system
- Tyre bead and pressure sensor
- Brake caliper/rotor thermal plume
- Upright and steering envelope
- Aero cover retention and pit procedure

CALCULATION FRAMEWORK

- Bending stress from radial/corner load combinations
- Torque path bearing stress at drive features
- Rotational inertia: $I = \int r^2 dm$
- Natural frequency separation from excitation orders

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloRail Center-Lock Forged Wheel - Materials, Manufacturing, and Quality

HW-06 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Wheel	Forged 6061-T6/6082 or higher-strength qualified alloy	Material/process must match validation route
Edge protector	Forged aluminum, polymer composite, or titanium	No tyre/bead interference
Aero cover	Carbon composite or high-temp polymer	Positive retention and brake-temperature validation
Inserts	Hardened wear inserts where needed	Galvanic isolation required

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Topology study followed by manufacturable forged geometry	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Flow-form/forging supplier DFM and grain-flow review	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	5-axis machining with controlled hub/seat datums	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Shot peen or approved finish without masking cracks	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	100% runout, balance, material lot, and NDT inspection	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloRail Center-Lock Forged Wheel - Verification, Failure Modes, and Release

HW-06 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Nonlinear FEA with tyre-load boundary sensitivity	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	SAE J2530-style radial/cornering fatigue and impact plan as applicable	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Rotary bending and torque fatigue at elevated temperature	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Wheel-to-wheel balance, runout, and CT/NDT sample inspections	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Track curb/endurance plus brake-temperature correlation	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Spoke or hub fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bead-seat cracking	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Drive pocket fretting	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Air cover release	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Heat-related property loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Tyre damage from protector/interface	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Documented load case and vehicle application
- ☐ Controlled runout and balance limits
- ☐ No interference under deflection/thermal growth
- ☐ Serialized retirement life
- ☐ Distinct inner/outer and left/right markings

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

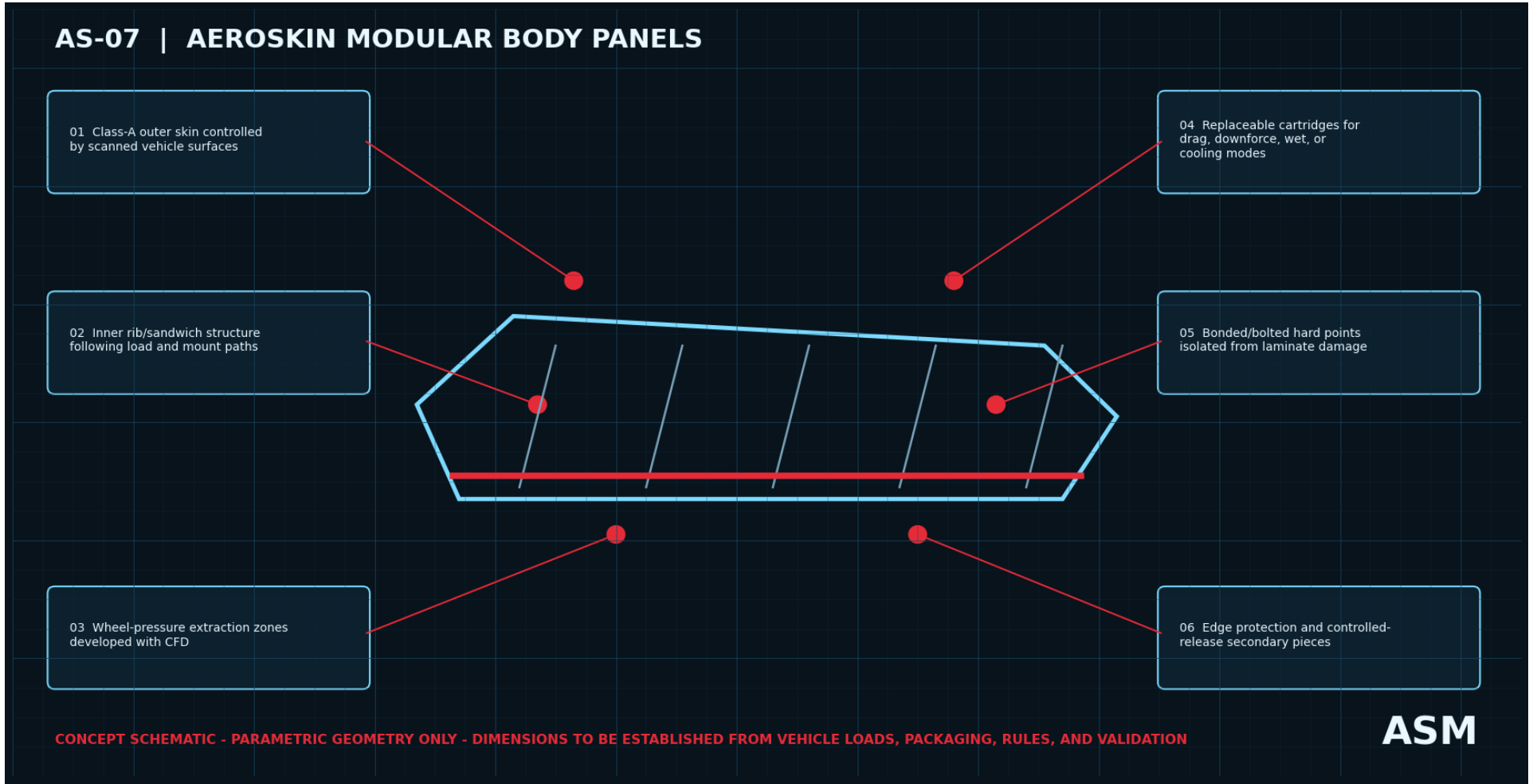
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

AeroSkin Modular Body Panels

AS-07 / SHEET 1 OF 4

AERO DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Build a repeatable family of lightweight exterior panels that combines aerodynamic function, fast service, and recognizable ASM surface language.	Internal Halo Spine ribs, curved Aero Vein extraction slots, modular vent cartridges, flush SaintKey fasteners, and sacrificial strike rails.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

AeroSkin Modular Body Panels - Architecture and Design Basis

AS-07 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Class-A outer skin controlled by scanned vehicle surfaces
- Inner rib/sandwich structure following load and mount paths
- Wheel-pressure extraction zones developed with CFD
- Replaceable cartridges for drag, downforce, wet, or cooling modes
- Bonded/bolted hard points isolated from laminate damage
- Edge protection and controlled-release secondary pieces

PRIMARY DESIGN VARIABLES

- External pressure and suction loads
- Panel flutter and local buckling
- Mount tolerance and body-gap control
- Heat exposure near brakes/exhaust
- Impact/stone/curb loading
- Repairability and maximum loose-part size after damage

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Chassis/body mounting datums
- Wheel/tyre and suspension travel
- Brake/cooling pressure zones
- Lights, sensors, tow points, and regulations
- Adjacent splitter/floor/wing components

CALCULATION FRAMEWORK

- Panel pressure load: $F = \text{integral}(p \text{ dA})$
- Sandwich bending stiffness scaling with face spacing
- Fastener pull-out and bearing allowables from coupon data
- Flutter margin from modal analysis and pressure excitation

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

AeroSkin Modular Body Panels - Materials, Manufacturing, and Quality

AS-07 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Skin	Pre-preg carbon/epoxy	Toughened resin and controlled cure
Impact layer	Aramid or hybrid carbon/aramid	Localized, not cosmetic only
Core	Nomex/aluminum honeycomb or structural foam	Closed edges and moisture control
Hard points	Titanium/aluminum inserts with isolation ply	Pull-out and galvanic validation

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Scan master vehicle and establish datum scheme	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	CFD surface iteration and packaging review	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Female mold/CNC tooling or printed prototype tools	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Ply-book, debulk, cure, trim fixture, and bonded inserts	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Dimensional fixture inspection and tap-test/ultrasound plan	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

AeroSkin Modular Body Panels - Verification, Failure Modes, and Release

AS-07 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	CFD plus wind-tunnel/track pressure-tap correlation	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Panel static proof and mount pull-out	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Vibration, heat soak, stone impact, water, and repair trials	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	High-speed flutter video/strain measurement	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Controlled damage test confirming no tyre or driver hazard	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Panel flutter or delamination	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Insert pull-out	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Vent cartridge release	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Heat damage	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Wheel contact	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Water ingress into core	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Master surface and tolerance map
- ☐ Approved ply book and cure record
- ☐ Positive secondary retention where required
- ☐ Field inspection/repair limits
- ☐ Configuration-controlled vent options

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

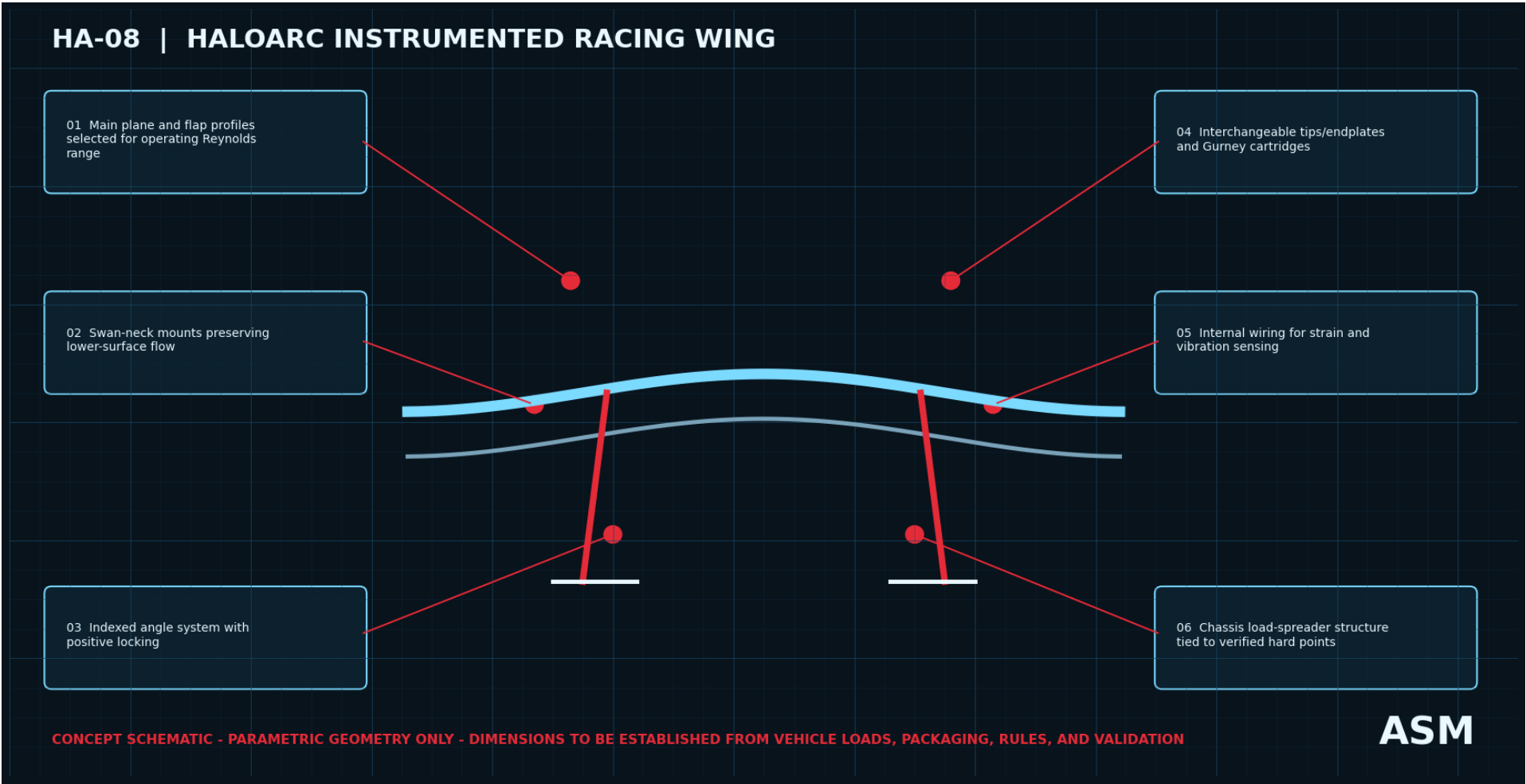
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

HaloArc Instrumented Racing Wing

HA-08 / SHEET 1 OF 4

AERO DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Create an adjustable rear wing whose geometry, mounts, and measured loads form a closed development loop rather than a cosmetic appendage.	Subtle central HaloArc platform, swan-neck carbon/titanium pylons, indexed HaloClick adjusters, replaceable vortex tips, and strain-measured load path.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloArc Instrumented Racing Wing - Architecture and Design Basis

HA-08 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Main plane and flap profiles selected for operating Reynolds range
- Swan-neck mounts preserving lower-surface flow
- Indexed angle system with positive locking
- Interchangeable tips/endplates and Gurney cartridges
- Internal wiring for strain and vibration sensing
- Chassis load-spreader structure tied to verified hard points

PRIMARY DESIGN VARIABLES

- Downforce, drag, balance, yaw sensitivity, and stall behavior
- Center of pressure migration with ride/yaw
- Wing deflection and twist under peak load
- Pylon buckling/fatigue and adjuster bearing stress
- Rear visibility and regulatory envelope
- Loss-of-fastener and impact behavior

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Deck/chassis hard points
- Rear body/floor wake
- Rules box and height/width limits
- Engine cover access
- Telemetry and power harness

CALCULATION FRAMEWORK

- Downforce: $F = 0.5 \rho V^2 S C_L$
- Drag: $D = 0.5 \rho V^2 S C_D$
- Bending moment: $M_{root} = \int y dF$
- Deflection/twist verified by FEA and proof test

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloArc Instrumented Racing Wing - Materials, Manufacturing, and Quality

HA-08 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Wing skins	Pre-preg carbon/epoxy	Autoclave or qualified OOA route
Core/spars	Foam/honeycomb plus carbon shear webs	Designed for torsion and local insert loads
Pylon core	Titanium or high-strength aluminum	Hidden inside aerodynamic fairing
Hardware	Titanium/high-strength steel	Positive locking and inspection

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	2D airfoil screening then 3D CFD	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Rapid prototype sections for tunnel/flow visualization	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Composite mold, spar mandrel, and bonded assembly development	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Proof-load fixture at multiple span stations	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Calibrated strain system correlated to known loads	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloArc Instrumented Racing Wing - Verification, Failure Modes, and Release

HA-08 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	CFD mesh/solver sensitivity and wind-tunnel correlation	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Static ultimate/proof load in multiple directions	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Fatigue spectrum, vibration, fastener loss, and adjuster wear	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Track aero map using load cells/strain, ride height, and pressure	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Debris/impact and safe degradation assessment	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Composite delamination	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Pylon buckling/fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Adjuster slip	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Tip/endplate release	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Unexpected stall or balance shift	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Chassis hard-point overload	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Published setup map by indexed position
- ☐ Measured load correlation band
- ☐ Proof-tested serial assembly
- ☐ Secondary locking and witness marks
- ☐ Defined damage and repair limits

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

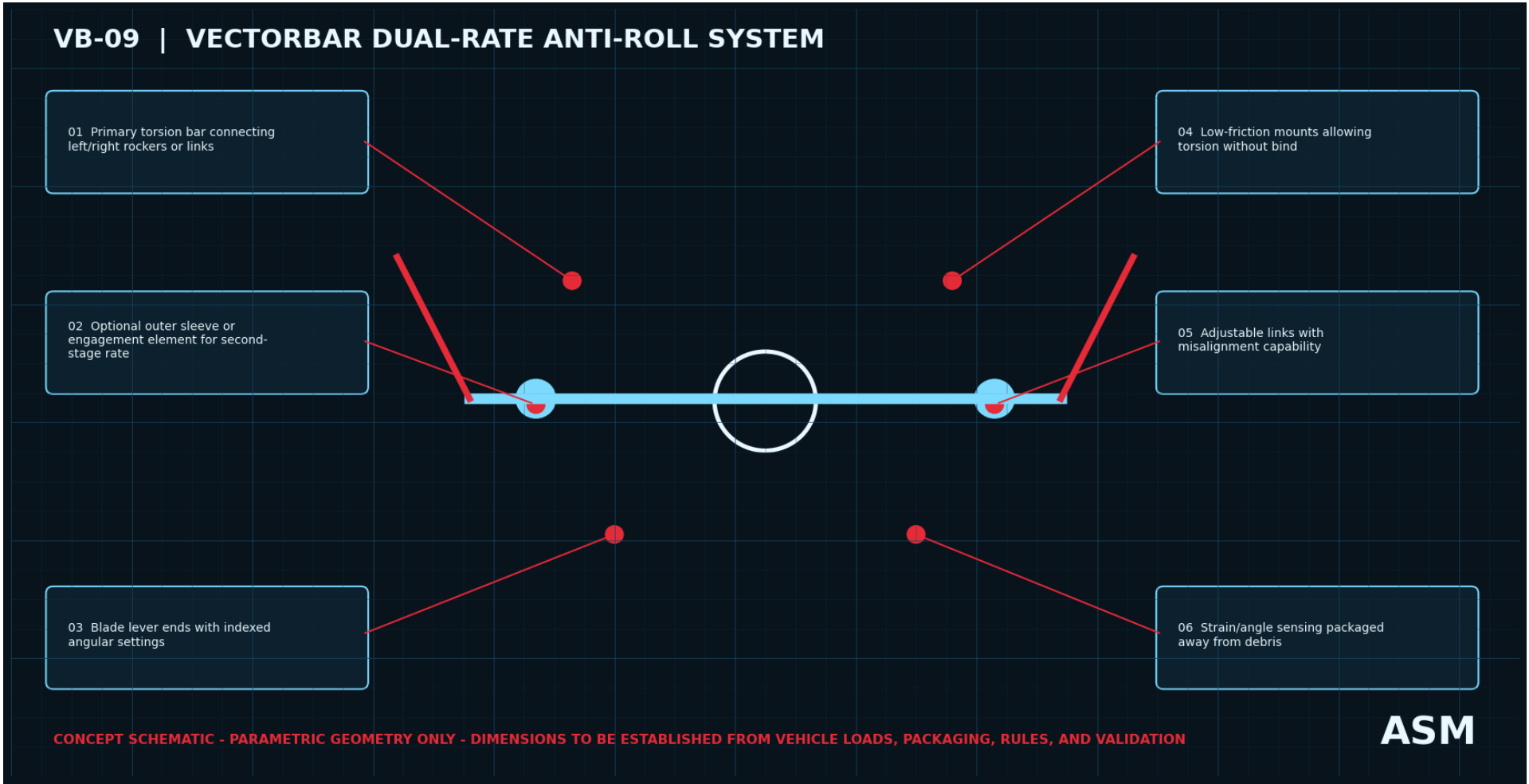
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

VectorBar Dual-Rate Anti-Roll System

VB-09 / SHEET 1 OF 4

CORNERING DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Provide fast, repeatable roll-stiffness tuning with a potential second-stage rate while retaining a clear fail-safe mechanical path.	Coaxial torsion architecture, HaloIndex blade adjusters, replaceable stiffness cartridges, and optional torsion-strain telemetry.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

VectorBar Dual-Rate Anti-Roll System - Architecture and Design Basis

VB-09 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Primary torsion bar connecting left/right rockers or links
- Optional outer sleeve or engagement element for second-stage rate
- Blade lever ends with indexed angular settings
- Low-friction mounts allowing torsion without bind
- Adjustable links with misalignment capability
- Strain/angle sensing packaged away from debris

PRIMARY DESIGN VARIABLES

- Wheel rate contribution through motion ratio
- Bar diameter/length/material and effective torsion length
- Blade section stiffness versus angle
- Engagement clearance and hysteresis for dual-rate concept
- Bearing friction, chassis mount stiffness, and link articulation
- Bump/roll cross-talk through actual suspension kinematics

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Suspension rockers/control arms
- Chassis mounts
- Ride-height and damper package
- Alignment/setup sheet
- Telemetry channel if instrumented

CALCULATION FRAMEWORK

- Solid bar: $k_t = GJ/L$
- Wheel roll stiffness from bar rate and motion ratios
- Blade stiffness from section properties and orientation
- Combined rate for staged engagement determined experimentally

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

VectorBar Dual-Rate Anti-Roll System - Materials, Manufacturing, and Quality

VB-09 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Bar/cartridge	Spring steel such as 300M/4340-family or qualified alternative	Heat treatment, shot peening, corrosion control
Blades	High-strength steel or titanium	Fatigue and notch sensitivity
Bearings	PTFE-lined spherical/plain bearings	Environment and life dependent
Housing	Aluminum or steel	Mount stiffness critical

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Straightness and spline machining before/after heat treatment as required	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Controlled heat treat, shot peen, coating, and hardness mapping	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Bench torsion calibration for every setting	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Kinematic fixture check for bind through full travel	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Serialized stiffness curve and NDT report	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

VectorBar Dual-Rate Anti-Roll System - Verification, Failure Modes, and Release

VB-09 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	FEA: torsion, blade bending, spline contact, mount loads	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Static calibration and hysteresis at every setting	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	High-cycle fatigue and overload/failure containment	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Contamination, corrosion, bearing wear, and link misalignment	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Track correlation versus roll gradient and tyre load distribution	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Torsion fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Spline fretting	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Blade slip/yield	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bearing seizure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Link buckling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Unexpected step engagement	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Repeatable indexed stiffness map
- ☐ Fail-safe primary load path
- ☐ No bind through kinematic envelope
- ☐ Defined overload/inspection limit
- ☐ Trackside change without unverified assembly

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

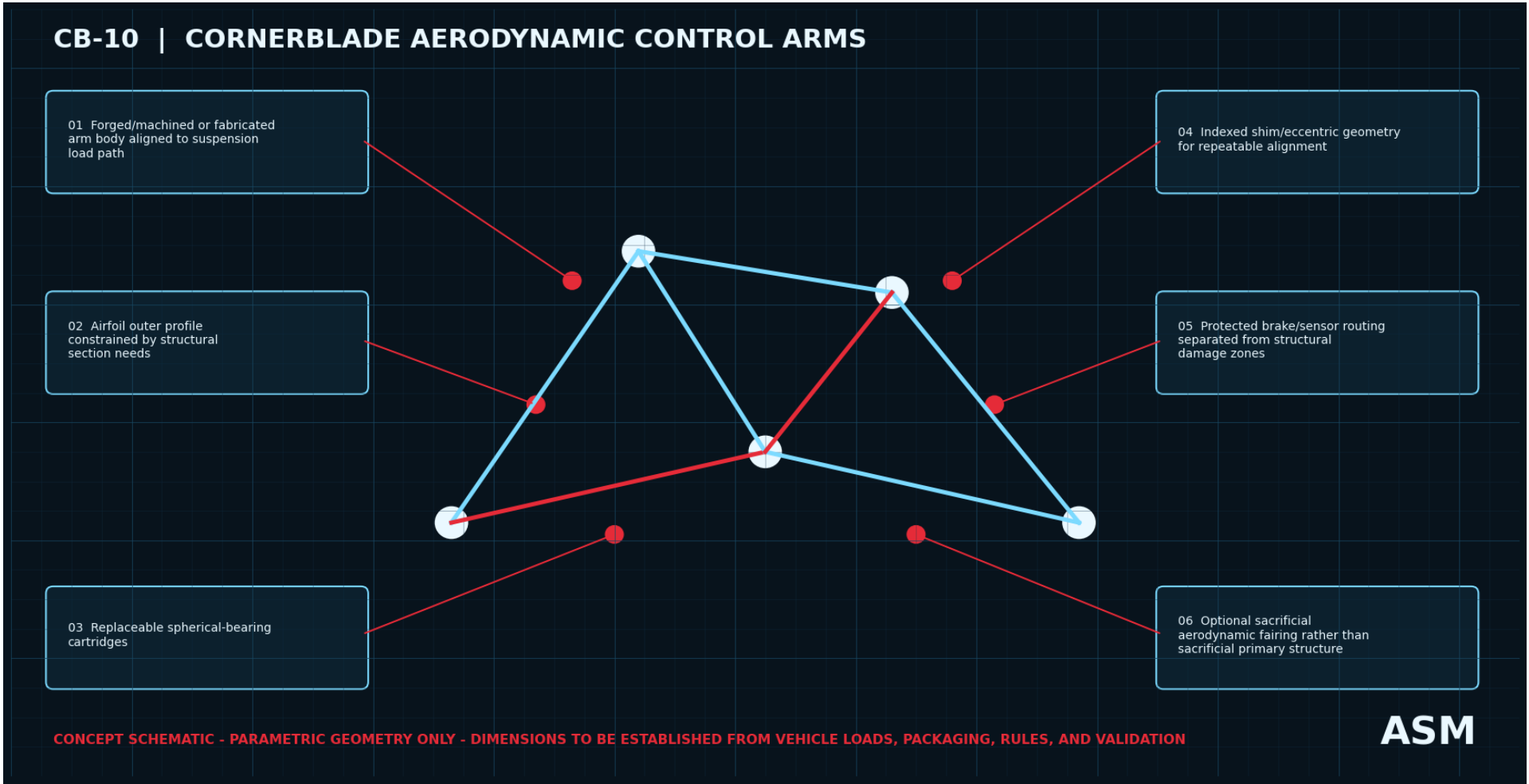
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

CornerBlade Aerodynamic Control Arms

CB-10 / SHEET 1 OF 4

CORNERING DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Reduce wheel-house drag and improve serviceability while preserving a transparent, inspectable suspension load path.	Airfoil arm sections, replaceable SaintLink bearing cartridges, indexed camber inserts, internal protected routing, and optional load sensing.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

CornerBlade Aerodynamic Control Arms - Architecture and Design Basis

CB-10 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Forged/machined or fabricated arm body aligned to suspension load path
- Airfoil outer profile constrained by structural section needs
- Replaceable spherical-bearing cartridges
- Indexed shim/eccentric geometry for repeatable alignment
- Protected brake/sensor routing separated from structural damage zones
- Optional sacrificial aerodynamic fairing rather than sacrificial primary structure

PRIMARY DESIGN VARIABLES

- Braking, cornering, vertical, curb, and combined loads
- Buckling, bending, bearing, and fatigue at all adjustment positions
- Joint articulation and misalignment
- Tyre/wheel/chassis clearance through kinematics
- Aerodynamic section angle across steer and ride
- Damage tolerance and crack inspection access

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Upright/knuckle
- Chassis/subframe pickups
- Spring/damper/ARB links
- Brake lines and sensors
- Wheel/tyre and bodywork

CALCULATION FRAMEWORK

- Axial stress: $\sigma = F/A$
- Buckling screen: $P_{cr} = \pi^2 E I / (K L)^2$
- Bearing stress and bolt shear at joints
- Fatigue from rainflow-counted vehicle load history

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

CornerBlade Aerodynamic Control Arms - Materials, Manufacturing, and Quality

CB-10 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Arm	Forged 7075/7050 aluminum, titanium, or qualified steel	Choose by packaging, fatigue, and cost
Bearing housing	Integral or replaceable hardened insert	Prevent fretting/ovalization
Fairing	Composite or thermoplastic	Nonstructural and replaceable
Fasteners	High-strength certified motorsport hardware	Controlled grip length and locking

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Vehicle load acquisition and multibody load extraction	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Topology/section study constrained by manufacturing	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Forging/billet prototype and 5-axis finishing	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Bearing press/retention development and dimensional fixture	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Proof, fatigue, NDT, and stiffness characterization	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

CornerBlade Aerodynamic Control Arms - Verification, Failure Modes, and Release

CB-10 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Nonlinear FEA at all alignment positions	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Static proof to defined load envelope	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Multiaxial fatigue using measured duty cycle	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Joint wear, contamination, corrosion, and curb impact	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Track strain correlation and post-event NDT	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Arm fatigue/buckling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bearing pull-out or seizure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Fastener loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Adjustment slip	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Hidden damage under fairing	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Line/harness chafe	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ No primary sacrificial fracture concept without full safety case
- ☐ Visible inspection access
- ☐ Positive adjustment retention
- ☐ Bearing cartridge life limits
- ☐ Proof-tested representative process

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

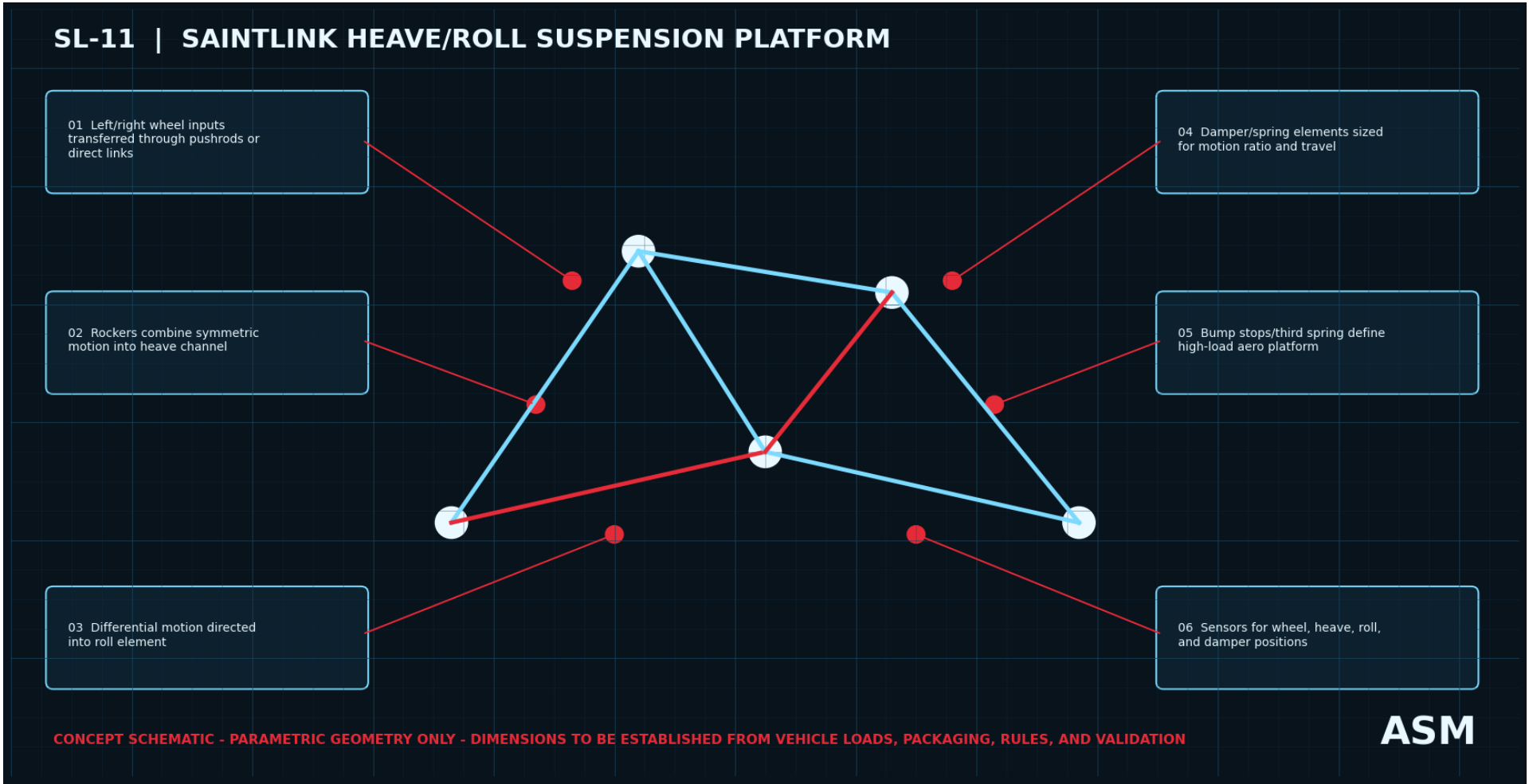
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

SaintLink Heave/Roll Suspension Platform

SL-11 / SHEET 1 OF 4

CORNERING DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Control heave, pitch, and roll more independently so aerodynamic ride height can be maintained without excessive single-wheel stiffness.	Central Halo heave element, separated roll path, indexed rockers, modular springs/dampers, and position/load sensing.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

SaintLink Heave/Roll Suspension Platform - Architecture and Design Basis

SL-11 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Left/right wheel inputs transferred through pushrods or direct links
- Rockers combine symmetric motion into heave channel
- Differential motion directed into roll element
- Damper/spring elements sized for motion ratio and travel
- Bump stops/third spring define high-load aero platform
- Sensors for wheel, heave, roll, and damper positions

PRIMARY DESIGN VARIABLES

- Kinematic motion ratios throughout travel
- Heave, roll, and pitch stiffness targets
- Damper velocity distribution and cavitation margin
- Aero load versus speed and ride height
- Bump-stop engagement and platform nonlinearity
- Packaging, service access, and single-point failure behavior

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Chassis pickups
- Uprights/control arms/pushrods
- Aero floor/wing loads
- Steering and driveshafts
- Ride-height and telemetry system

CALCULATION FRAMEWORK

- Wheel rate = spring rate times motion ratio squared
- Heave/roll coordinates formed from left/right wheel motions
- Damper force from measured F-v curve
- Natural frequency and damping ratio as system-level screens

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

SaintLink Heave/Roll Suspension Platform - Materials, Manufacturing, and Quality

SL-11 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Rockers	7075 aluminum or titanium	Bearing and fatigue critical
Links	Steel/titanium/aluminum by load and buckling	Rod-end articulation controlled
Springs	Motorsport coil/Belleville/torsion elements	Supplier-qualified rates
Dampers	Serviceable motorsport units	Thermal and force-velocity mapping

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Build multibody kinematic model from surveyed hard points	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Rig prototype with adjustable pickup fixture	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Machine rockers and brackets from low-cost mule material first	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Instrumented spring/damper rig characterization	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Corner-rig and seven-post correlation before high-speed release	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

SaintLink Heave/Roll Suspension Platform - Verification, Failure Modes, and Release

SL-11 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Kinematic sweep and tolerance Monte Carlo	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Static proof and fatigue of rockers/links/mounts	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Damper dyno and thermal endurance	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	K&C;/corner rig for compliance and mode separation	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Track correlation of ride height, load, and tyre temperature	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Rocker/link fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bearing seizure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Damper overheating	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bump-stop instability	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Mode cross-coupling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Sensor-driven setup error	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Stable, monotonic kinematics
- ☐ Defined mechanical fallback if sensor fails
- ☐ No interference at full travel/steer
- ☐ Setup map tied to measured wheel rates
- ☐ Documented hard-point loads

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

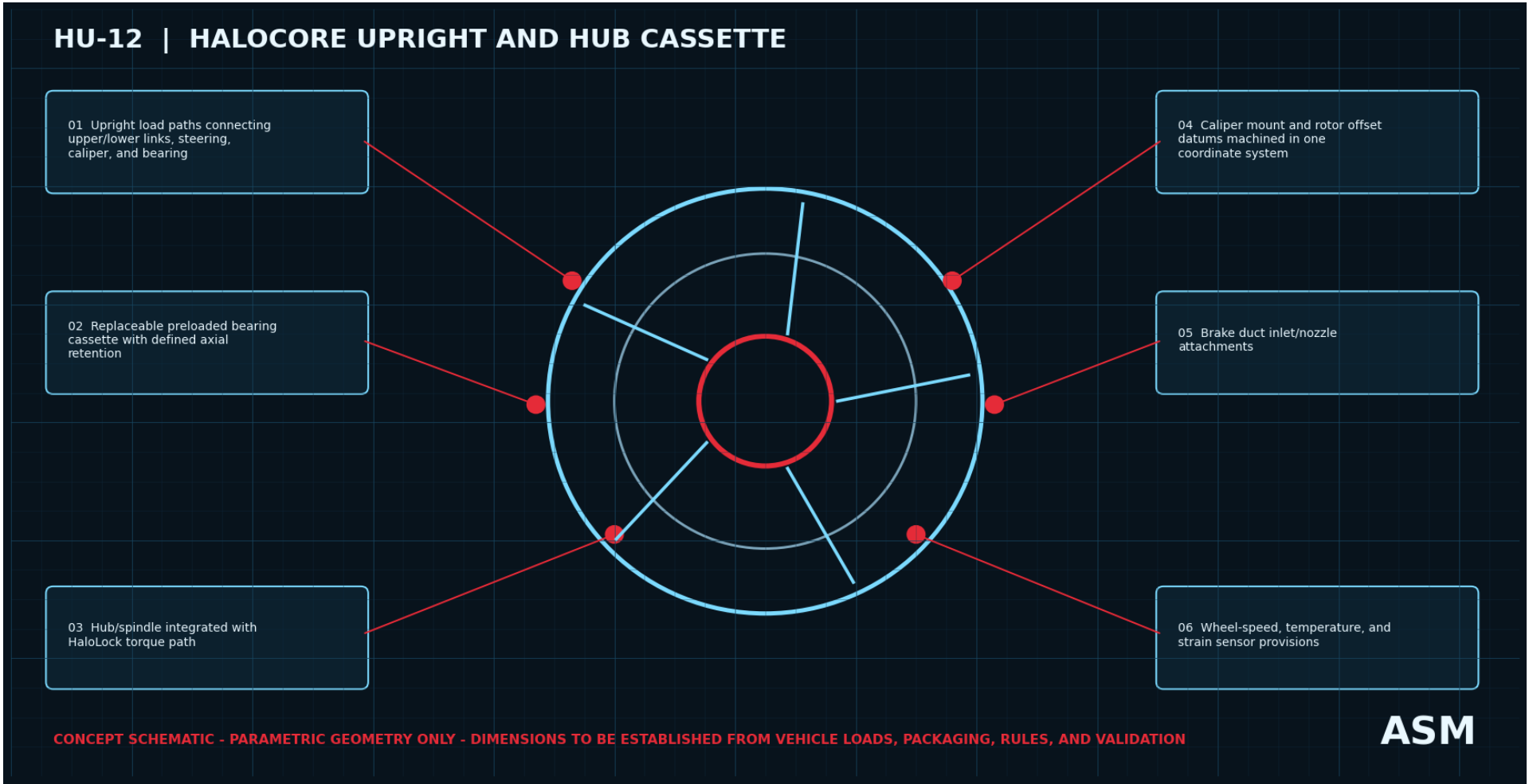
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

HaloCore Upright and Hub Cassette

HU-12 / SHEET 1 OF 4

WHEEL DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Integrate wheel bearing, center-lock, brake mounting, ducting, and sensing into a stiff, quickly serviced corner module.	Topology-shaped HaloCore, keyed bearing cassette, integrated duct passages, protected sensor spine, and inspection windows.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloCore Upright and Hub Cassette - Architecture and Design Basis

HU-12 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Upright load paths connecting upper/lower links, steering, caliper, and bearing
- Replaceable preloaded bearing cassette with defined axial retention
- Hub/spindle integrated with HaloLock torque path
- Caliper mount and rotor offset datums machined in one coordinate system
- Brake duct inlet/nozzle attachments
- Wheel-speed, temperature, and strain sensor provisions

PRIMARY DESIGN VARIABLES

- Combined wheel loads and braking/drive torque
- Bearing life, preload, stiffness, and temperature
- Steering geometry and compliance steer
- Caliper mount stiffness and rotor runout stack
- Fastener access, pit replacement, and contamination sealing
- Unsprung mass versus fatigue/damage tolerance

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Control arms/tie rod
- Wheel/center lock
- Rotor/caliper/duct
- Driveshaft where driven
- Sensors/telemetry

CALCULATION FRAMEWORK

- Bearing equivalent load and life per supplier method
- Upright compliance from FEA and K&C; targets
- Bolt-group/caliper torque reaction
- Thermal fit change from differential expansion

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloCore Upright and Hub Cassette - Materials, Manufacturing, and Quality

HU-12 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Upright	Forged 7075/7050 aluminum, titanium, or cast/forged steel	Application and validation dependent
Hub/spindle	High-strength alloy steel or titanium	Fatigue, wear, and NDT critical
Bearing	Motorsport angular-contact/tapered cassette	Supplier load/speed/temperature data
Duct shields	Composite/aluminum/stainless	Replaceable

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Multibody load extraction and topology optimization	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Forging/casting DFM followed by datum-driven machining	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Bearing fit/preload process with thermal assembly study	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	CMM, surface finish, hardness, and NDT inspection	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Corner-module proof rig including caliper and wheel loads	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloCore Upright and Hub Cassette - Verification, Failure Modes, and Release

HU-12 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	FEA load matrix and modal analysis	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Static proof of all load directions and brake torque	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Bearing endurance/temperature/seal contamination	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Corner fatigue spectrum with representative joints	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Track strain/temp correlation and post-test NDT	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Upright fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Bearing overheat or preload loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Hub/spindle crack	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Caliper mount deflection	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Sensor damage	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Cassette retention failure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Serviceable cassette without datum loss
- ☐ Defined bearing replacement triggers
- ☐ Controlled rotor/wheel runout stack
- ☐ NDT-accessible high-stress zones
- ☐ Life tracking by corner

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

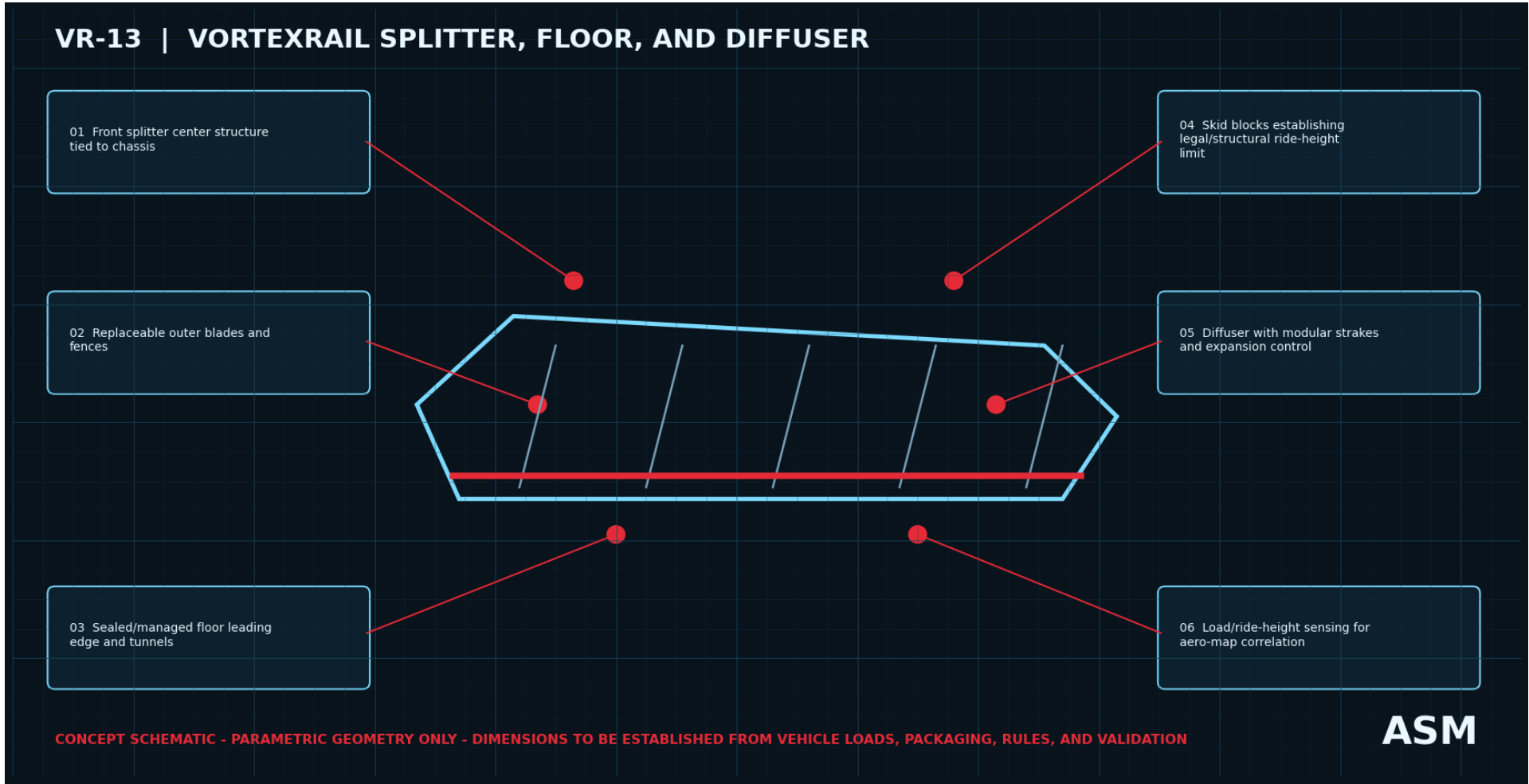
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

VortexRail Splitter, Floor, and Diffuser

VR-13 / SHEET 1 OF 4

AERO DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Create a modular underbody package that produces stable downforce, protects expensive structure, and can be tuned by circuit.	VortexRail edge geometry, load-sensing splitter stays, replaceable strake cartridges, sacrificial skid rails, and controlled-release outer blades.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

VortexRail Splitter, Floor, and Diffuser - Architecture and Design Basis

VR-13 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Front splitter center structure tied to chassis
- Replaceable outer blades and fences
- Sealed/managed floor leading edge and tunnels
- Skid blocks establishing legal/structural ride-height limit
- Diffuser with modular strakes and expansion control
- Load/ride-height sensing for aero-map correlation

PRIMARY DESIGN VARIABLES

- Downforce/drag and center-of-pressure map versus ride, pitch, yaw
- Ground clearance and stall sensitivity
- Splitter bending and support-rod load
- Floor suction, panel buckling, and sealing
- Curb strike, abrasion, debris, and heat exposure
- Safe release/retention of damaged outer pieces

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Chassis/front crash structure
- Body panels and wheel wakes
- Cooling inlets/exits
- Exhaust and heat shields
- Regulation reference plane

CALCULATION FRAMEWORK

- Aero forces: $0.5 \rho V^2 S C$
- Panel pressure integration and support reactions
- Beam/plate deflection for splitter/floor
- Wear rate versus contact pressure and speed

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

VortexRail Splitter, Floor, and Diffuser - Materials, Manufacturing, and Quality

VR-13 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Primary structure	Carbon sandwich with toughened resin	High stiffness and controlled core closeouts
Skids	Titanium, hardwood/composite, or regulation-approved material	Wear and heat dependent
Outer blades	Carbon/aramid or thermoplastic composite	Replaceable
Stays	Steel/titanium with calibrated load cells optional	Buckling and rod-end control

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	CFD ride/yaw matrix and geometry DOE	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Scale model or full-scale tunnel where feasible	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Flat tooling, bonded core/inserts, CNC trim fixture	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Static load fixture and abrasion/strike coupons	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Track rake, pressure tap, and load measurement	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

VortexRail Splitter, Floor, and Diffuser - Verification, Failure Modes, and Release

VR-13 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	CFD verification and tunnel/track correlation	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Static proof and fatigue of stays/mounts	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Curb strike/drop/abrasion and heat tests	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	High-speed ride-height sweep with safe limits	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Damage containment and loose-part assessment	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Ground-effect stall	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Splitter stay buckling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Panel delamination	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Skid wear-through	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Strake release	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Chassis mount overload	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Defined safe ride-height window
- ☐ Measured aero map by configuration
- ☐ Replaceable damage zones
- ☐ Wear gauges and inspection intervals
- ☐ No unretained sharp debris

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

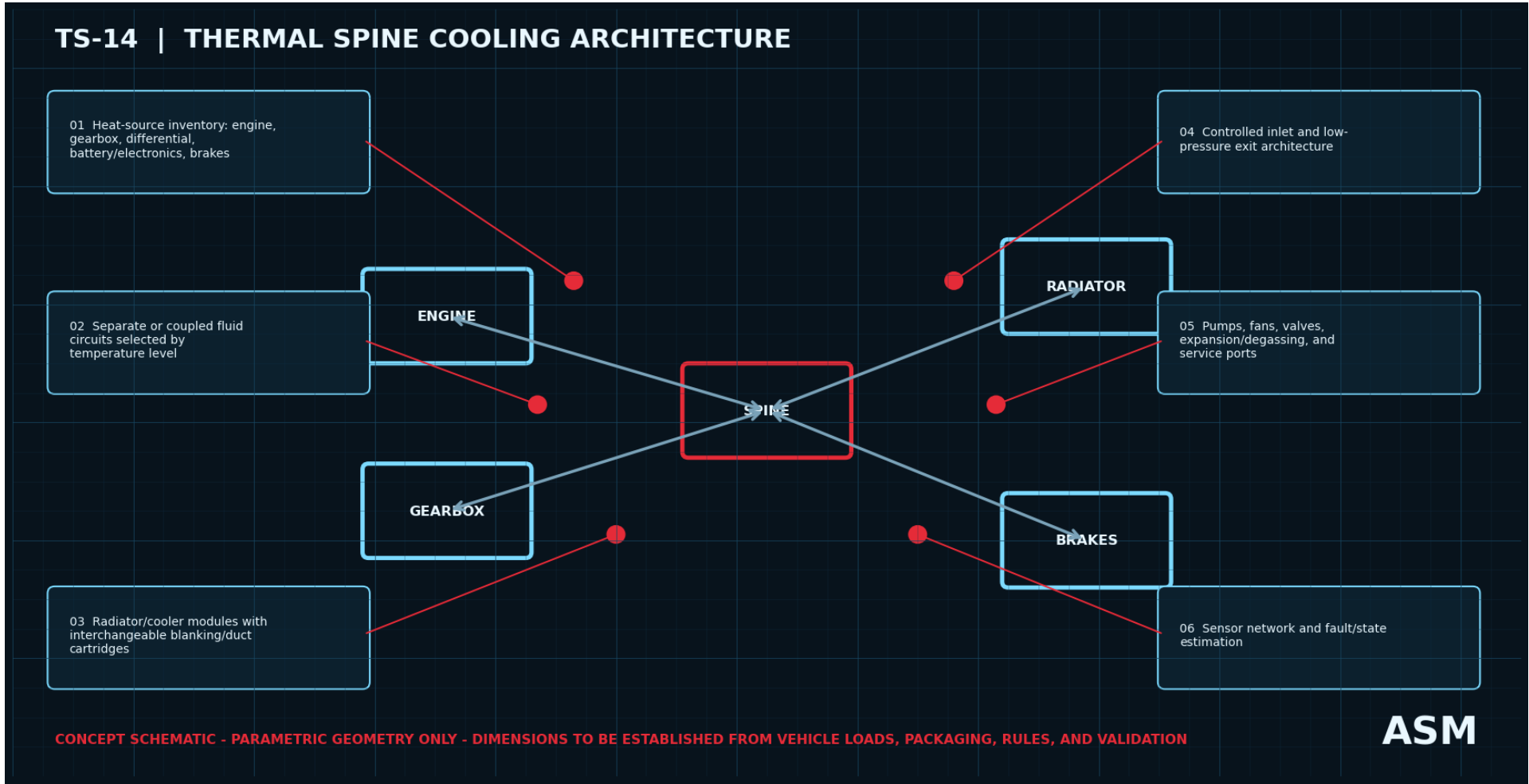
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

Thermal Spine Cooling Architecture

TS-14 / SHEET 1 OF 4

THERMAL DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Manage all vehicle heat rejection as one controlled system, delivering only the airflow and pump power needed for each session.	Central thermal backbone, modular cooling cartridges, Aero Vein exits, pressure/temperature sensing, and Sentinel setup recommendations.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

Thermal Spine Cooling Architecture - Architecture and Design Basis

TS-14 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Heat-source inventory: engine, gearbox, differential, battery/electronics, brakes
- Separate or coupled fluid circuits selected by temperature level
- Radiator/cooler modules with interchangeable blanking/duct cartridges
- Controlled inlet and low-pressure exit architecture
- Pumps, fans, valves, expansion/degassing, and service ports
- Sensor network and fault/state estimation

PRIMARY DESIGN VARIABLES

- Steady and transient heat rejection by operating case
- Coolant/oil flow, pressure drop, cavitation, and boiling margin
- Radiator effectiveness and air-side pressure loss
- Thermal soak after shutdown
- Cross-contamination and leak containment
- Drag/power penalty versus temperature control

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Powertrain and brake systems
- Aero inlets/exits
- Electrical power and control
- Firewall/cabin safety
- Service/fill/bleed equipment

CALCULATION FRAMEWORK

- Heat balance: $\dot{Q} = \dot{m} c_p \Delta T$
- Heat exchanger effectiveness method
- Pump power = $\Delta P Q / \text{efficiency}$
- Transient storage: $Q = m c_p \Delta T$

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

Thermal Spine Cooling Architecture - Materials, Manufacturing, and Quality

TS-14 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Heat exchangers	Aluminum motorsport cores or qualified supplier modules	Pressure/temperature/fatigue validated
Lines	Aluminum hardline and PTFE/qualified hose	Fire, abrasion, and pressure controls
Ducts	Carbon/high-temp thermoplastic	Serviceable modules
Valves/pumps	Automotive/motorsport-qualified	Diagnostics required

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Create thermal network model and duty cycles	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Bench characterize pumps, valves, radiators, and duct pressure loss	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Prototype modular rig with transparent/serviceable sections	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Pressure/leak/thermal-shock and vibration testing	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Instrumented vehicle heat-balance correlation	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

Thermal Spine Cooling Architecture - Verification, Failure Modes, and Release

TS-14 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Component bench performance maps	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Pressure proof/burst and leak decay	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Thermal shock, vibration, contamination, and hose aging	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Hot-climate, low-speed, high-speed, and shutdown soak vehicle tests	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Fault injection for sensor, pump, fan, and valve failures	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Coolant/oil leak	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Cavitation or air lock	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Radiator blockage	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Pump/fan failure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Overcooling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Heat-soak damage	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Documented thermal budget and margins
- ☐ Bleed/fill repeatability
- ☐ Safe degraded modes
- ☐ Measured drag/temperature trade-off
- ☐ Service intervals and fluid compatibility

RELEASE HOLD POINTS

1. Independent calculation/model review.
2. Drawing and tolerance review.
3. Supplier process-readiness review.
4. Design validation report approved.
5. Production validation on representative parts.
6. Vehicle integration and rules/compliance review.
7. Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

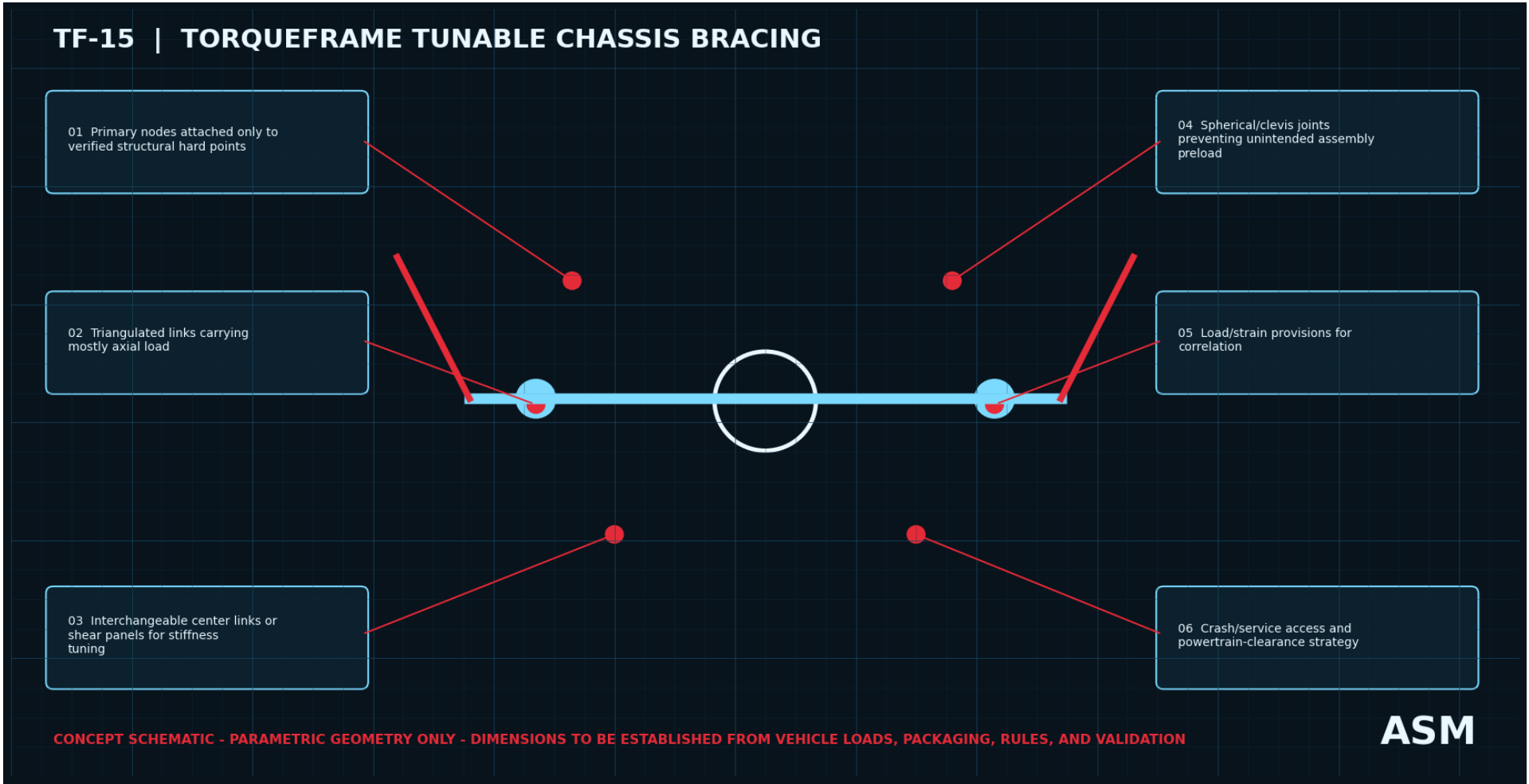
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

TorqueFrame Tunable Chassis Bracing

TF-15 / SHEET 1 OF 4

STRUCTURAL DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Provide controlled chassis stiffness changes using a measured, modular brace system rather than cosmetic bars.	Halo-shaped node plates, replaceable stiffness links, strain-marked load paths, and serialized setup modules.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

TorqueFrame Tunable Chassis Bracing - Architecture and Design Basis

TF-15 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Primary nodes attached only to verified structural hard points
- Triangulated links carrying mostly axial load
- Interchangeable center links or shear panels for stiffness tuning
- Spherical/clevis joints preventing unintended assembly preload
- Load/strain provisions for correlation
- Crash/service access and powertrain-clearance strategy

PRIMARY DESIGN VARIABLES

- Target torsional/bending stiffness change
- Local body/subframe load introduction
- Link buckling and joint fatigue
- Noise/vibration path changes
- Thermal growth and assembly tolerance
- Crash compatibility and serviceability

INTERFACE CONTROL

- Chassis/subframe hard points
- Powertrain/exhaust
- Suspension loads
- Crash structures
- Service/removal sequence

CALCULATION FRAMEWORK

- Axial link stiffness: $k = EA/L$
- Euler buckling screen
- Joint slip/preload analysis
- Vehicle torsional stiffness from applied moment/angle

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

TorqueFrame Tunable Chassis Bracing - Materials, Manufacturing, and Quality

TF-15 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Links	Thin-wall high-strength steel, aluminum, or carbon tube	Buckling, joint bonding/welding, inspection
Nodes	Machined/formed high-strength alloy	Load-spreading and bearing
Bolts	Certified high-strength hardware	Grip and preload control
Bushings/joints	Rigid or isolated by objective	Do not mix without model

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Baseline chassis torsion/compliance measurement	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	FEA and topology of candidate load paths	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Fabricated mule with strain gauges	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Controlled weld/bond procedures and alignment fixture	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Torsion rig plus durability and NVH assessment	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

TorqueFrame Tunable Chassis Bracing - Verification, Failure Modes, and Release

TF-15 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Chassis torsion rig with and without each module	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Static proof and fatigue of links/nodes	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Fastener preload retention and fretting	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Road/track durability and NVH measurement	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Crash-system review by qualified engineer	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Local chassis cracking	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Link buckling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Joint slip/fretting	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Unwanted load bypass	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
NVH increase	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Crash incompatibility	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Measured stiffness delta per module
- ☐ No attachment to unverified sheet metal
- ☐ Positive setup ID
- ☐ Defined inspection around nodes
- ☐ No claim of handling benefit without vehicle data

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

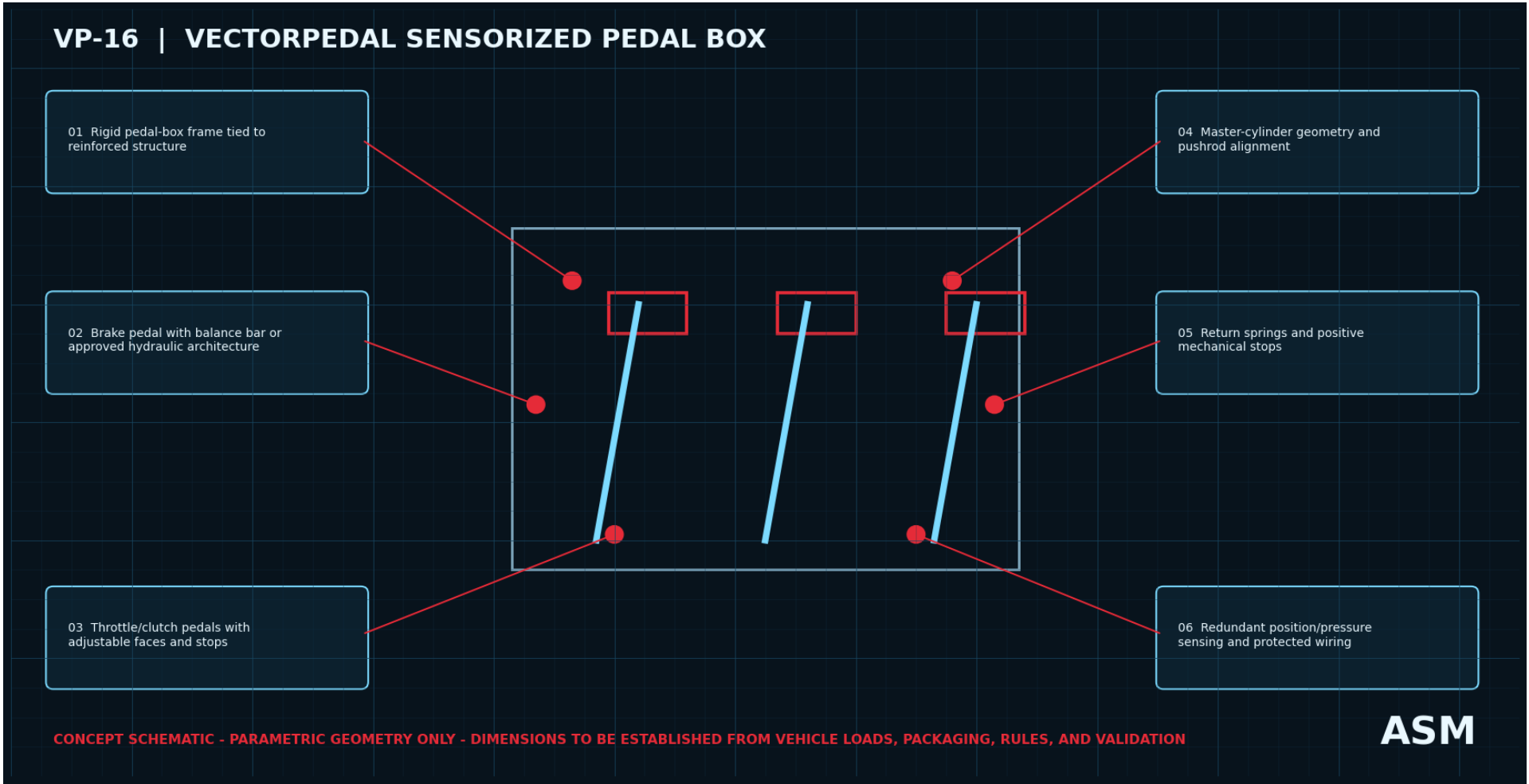
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

VectorPedal Sensorized Pedal Box

VP-16 / SHEET 1 OF 4

DRIVER DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Give drivers repeatable ergonomics and engineers high-quality pressure/position data while maintaining robust mechanical controls.	Trinity pedal faces, indexed ratio inserts, captured balance-bar settings, modular heel plate, and dual-sensor channels.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

VectorPedal Sensorized Pedal Box - Architecture and Design Basis

VP-16 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Rigid pedal-box frame tied to reinforced structure
- Brake pedal with balance bar or approved hydraulic architecture
- Throttle/clutch pedals with adjustable faces and stops
- Master-cylinder geometry and pushrod alignment
- Return springs and positive mechanical stops
- Redundant position/pressure sensing and protected wiring

PRIMARY DESIGN VARIABLES

- Pedal ratio, travel, force, and driver anthropometrics
- Master-cylinder bore selection and hydraulic balance
- Buckling/misalignment of pushrods
- Frame deflection and firewall/floor stiffness
- Sensor plausibility and failure response
- Foot entrapment, sharp edges, and emergency egress

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Master cylinders/brake lines
- Driver seat/harness/steering
- Firewall/floor structure
- ECU/data logger
- Throttle and clutch actuation

CALCULATION FRAMEWORK

- Pedal force = hydraulic force / pedal ratio adjusted for losses
- Line pressure = pedal force times ratio / master area
- Balance-bar front/rear pressure relationship
- Frame stiffness from force/deflection

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

VectorPedal Sensorized Pedal Box - Materials, Manufacturing, and Quality

VP-16 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Frame/pedals	7075/6061 aluminum, steel, or mixed construction	Fatigue and stiffness validated
Pins/shafts	Hardened steel with bushings/bearings	Wear and retention
Pedal faces	Aluminum with bonded/mechanical grip	No loose inserts
Sensors	Motorsport-qualified dual-channel	Environmental protection

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Ergonomic buck and driver fit sessions	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	FEA and bench stiffness proof	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Precision jig for pivot/master alignment	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Hydraulic pressure/leak test and sensor calibration	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Vehicle static/rolling brake commissioning	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

VectorPedal Sensorized Pedal Box - Verification, Failure Modes, and Release

VP-16 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Static overload and fatigue cycling	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Master/pushrod misalignment and jam checks	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Sensor plausibility/fault injection	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Brake balance adjustment retention	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Driver fit, heel-toe, wet footwear, and egress testing	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Pivot or frame fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Pushrod bind/buckling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Balance-bar migration	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Return spring loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Sensor disagreement	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Foot slip/entrapment	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Positive mechanical stops
- ☐ Redundant return strategy where appropriate
- ☐ Locked and witnessed balance adjustment
- ☐ Calibrated force/pressure map
- ☐ Driver-specific setup record

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

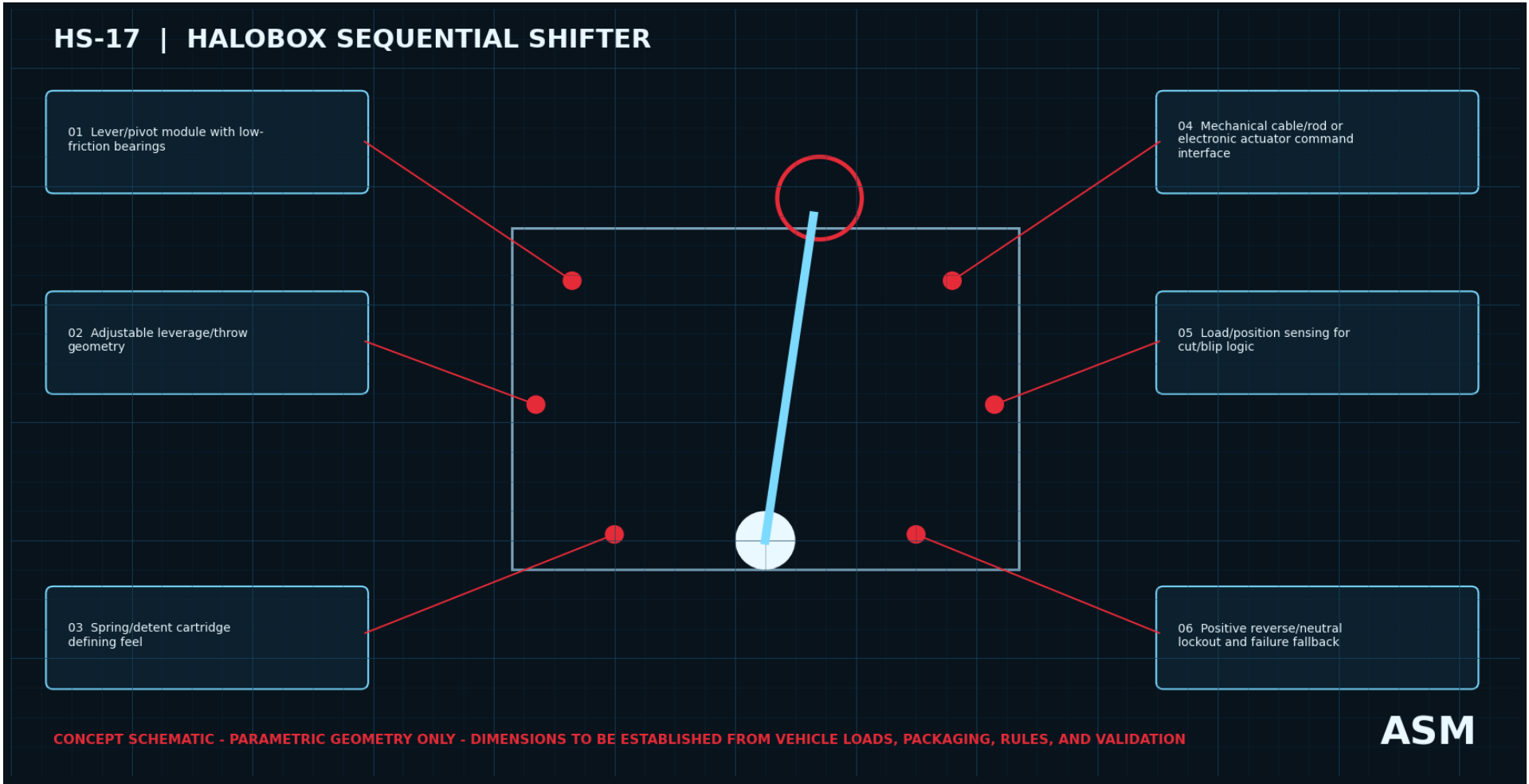
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

HaloBox Sequential Shifter

HS-17 / SHEET 1 OF 4

DRIVER DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Create a configurable, sensorized shift interface with clean detent feel and no ambiguity between driver input and gearbox actuation.	Halo lever arch, adjustable force cartridge, indexed throw positions, load-cell handle, and protected reverse/neutral logic.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

HaloBox Sequential Shifter - Architecture and Design Basis

HS-17 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Lever/pivot module with low-friction bearings
- Adjustable leverage/throw geometry
- Spring/detent cartridge defining feel
- Mechanical cable/rod or electronic actuator command interface
- Load/position sensing for cut/blip logic
- Positive reverse/neutral lockout and failure fallback

PRIMARY DESIGN VARIABLES

- Required shift force/throw and gearbox input travel
- Pivot/lever fatigue and driver abuse loads
- Cable/rod buckling, backlash, and friction
- Sensor latency/plausibility
- Misshift prevention and lockout ergonomics
- Mount stiffness and cockpit vibration

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Gearbox selector or actuator
- ECU ignition/fuel cut
- Driver cockpit/seat/harness
- Telemetry
- Reverse safety circuit

CALCULATION FRAMEWORK

- Lever ratio from input/output arm lengths
- Work/energy through detent and gearbox mechanism
- Buckling screen for rods
- Sensor threshold and timing derived from measured force trace

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

HaloBox Sequential Shifter - Materials, Manufacturing, and Quality

HS-17 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Lever/frame	Aluminum, steel, or titanium by load	Fatigue and stiffness
Pivot	Hardened shaft with rolling/plain bearing	Low play and contamination control
Handle	Composite/aluminum with load cell option	Ergonomic and thermal
Springs/detents	Spring steel	Life characterized

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Bench rig reproducing gearbox input force curve	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Adjustable mule and driver evaluation	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	CNC/fabricated housing with controlled pivot alignment	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Cycle endurance and environmental test	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Vehicle calibration with cut/blip controller	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

HaloBox Sequential Shifter - Verification, Failure Modes, and Release

HS-17 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Static abuse and million-cycle-type development target as appropriate	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Backlash/wear/contamination/temperature	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Sensor fault injection and missed-shift logging	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Driver trials with gloves and vibration	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Vehicle shift-quality and gearbox-dog inspection	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Lever/pivot fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Cable/rod jam	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
False sensor trigger	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Detent wear	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Reverse lockout failure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Mount loosening	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ No single sensor required for mechanical gear retention
- ☐ Positive lockouts
- ☐ Documented force/throw settings
- ☐ Serviceable wear cartridge
- ☐ Shift event traceability

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

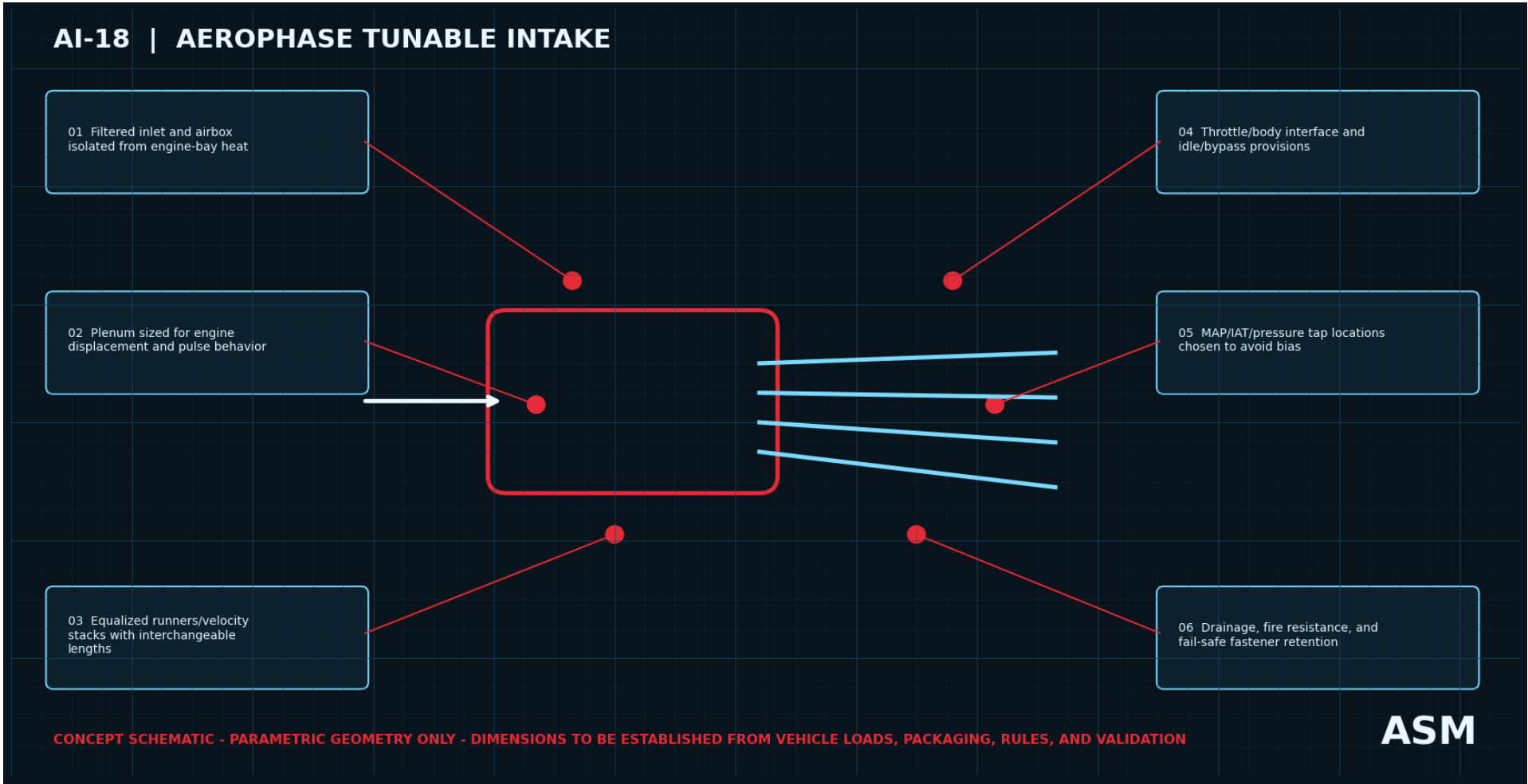
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

AeroPhase Tunable Intake

AI-18 / SHEET 1 OF 4

POWERTRAIN DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Tune engine breathing by interchangeable runner and plenum modules while preserving filtration, sealing, and calibration control.	Halo plenum rib, numbered velocity-stack cartridges, equal-path Aero Veins, and integrated pressure/temperature instrumentation.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

AeroPhase Tunable Intake - Architecture and Design Basis

AI-18 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Filtered inlet and airbox isolated from engine-bay heat
- Plenum sized for engine displacement and pulse behavior
- Equalized runners/velocity stacks with interchangeable lengths
- Throttle/body interface and idle/bypass provisions
- MAP/IAT/pressure tap locations chosen to avoid bias
- Drainage, fire resistance, and fail-safe fastener retention

PRIMARY DESIGN VARIABLES

- Air demand, pressure loss, and distribution cylinder-to-cylinder
- Runner length/area and resonance bands
- Plenum volume and throttle transient response
- Heat soak and filtration efficiency
- Structural vacuum/backfire/boost loads where applicable
- ECU recalibration for every hardware configuration

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Throttle/engine ports
- Air filter/snorkel/bodywork
- Fuel injectors/rails
- ECU sensors and calibration
- Fire system/heat shields

CALCULATION FRAMEWORK

- Helmholtz/runner resonance used only as initial models
- Mass flow and pressure loss from bench data
- Volumetric efficiency and torque measured on dyno
- Structural pressure/vacuum load cases

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

AeroPhase Tunable Intake - Materials, Manufacturing, and Quality

AI-18 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Plenum/runners	Carbon composite, cast/machined aluminum, or high-temp polymer	Fuel/heat/backfire compatibility
Stacks	Machined/printed qualified polymer or aluminum	Surface finish and repeatability
Seals	Fuel/oil/temperature-compatible elastomer	No ingestion risk
Fasteners	Captive and positively retained	Engine ingestion prevention

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	1D engine simulation and steady-flow bench	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Rapid prototype plenum/runner DOE	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Composite or machined production tooling	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Leak, vacuum/pressure, heat, and vibration tests	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Engine dyno calibration and cylinder-balance measurement	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

AeroPhase Tunable Intake - Verification, Failure Modes, and Release

AI-18 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Flow bench distribution and pressure loss	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Pressure/vacuum proof and backfire scenario review	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Thermal/vibration/fuel-oil exposure	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Engine dyno torque, AFR, knock, and cylinder balance	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Track heat-soak and filter contamination	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Air leak or seal ingestion	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Runner/fastener ingestion	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Unequal cylinder filling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Heat soak power loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Backfire damage	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Calibration mismatch	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Captive hardware
- ☐ Configuration-linked calibration ID
- ☐ Leak-tested serialized assembly
- ☐ Published filter/service interval
- ☐ No dyno claim without corrected data

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

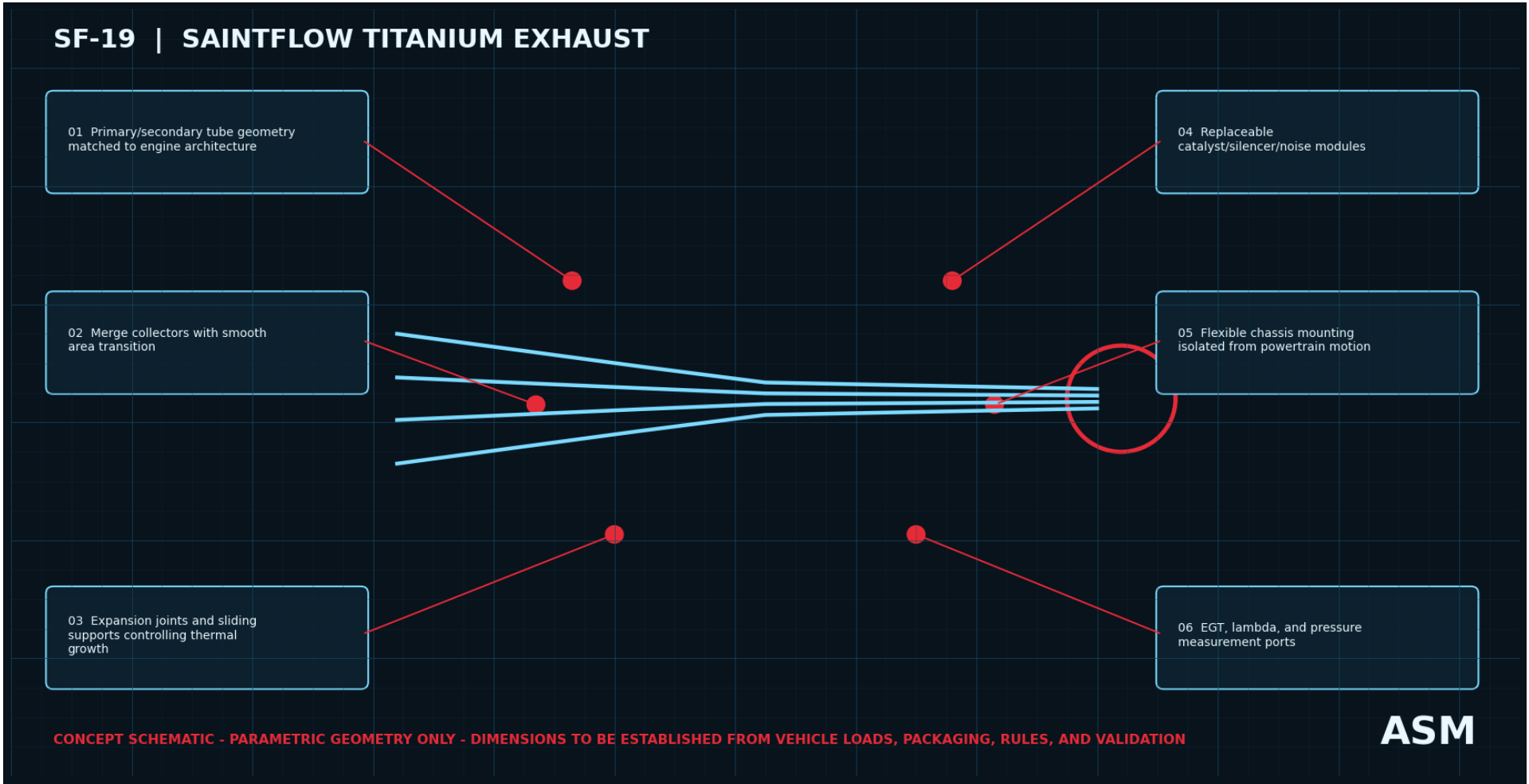
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

SaintFlow Titanium Exhaust

SF-19 / SHEET 1 OF 4

POWERTRAIN DYNAMICS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Reduce mass and manage pulse/heat with modular, repairable sections that are recognizable without compromising reliability or rules compliance.	Equal-path collectors, Halo expansion joints, modular silencer/catalyst sections, three-element outlet motif, and integrated temperature/pressure ports.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

SaintFlow Titanium Exhaust - Architecture and Design Basis

SF-19 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Primary/secondary tube geometry matched to engine architecture
- Merge collectors with smooth area transition
- Expansion joints and sliding supports controlling thermal growth
- Replaceable catalyst/silencer/noise modules
- Flexible chassis mounting isolated from powertrain motion
- EGT, lambda, and pressure measurement ports

PRIMARY DESIGN VARIABLES

- Gas temperature/flow and pressure drop
- Primary/collector tuning range
- Thermal expansion and mount reaction
- Tube/silencer fatigue under vibration
- Heat shielding and nearby component temperatures
- Noise/emissions/rule compliance by market/event

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Cylinder head/turbo/catalyst
- Body/floor/diffuser
- Mounts/powertrain motion
- Sensors/ECU
- Noise/emissions requirements

CALCULATION FRAMEWORK

- Thermal growth: $\Delta L = \alpha L \Delta T$
- Pressure loss from measured flow rather than simple diameter claims
- Tube/collector thermal stress and vibration fatigue
- Acoustic level measured to applicable procedure

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

SaintFlow Titanium Exhaust - Materials, Manufacturing, and Quality

SF-19 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Hot tubes	Grade 2/9 titanium, Inconel, or stainless by temperature	Material selected by measured temperature
Collectors	Formed/fabricated matching alloy	Weld quality critical
Silencers	Titanium/stainless shell with qualified packing	Fire and durability
Shields	Inconel/foil/composite systems	Air gap and attachment control

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Engine gas model and packaging CAD	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Purge-welded prototype with weld coupons	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Fixture control for flange/collector alignment	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Leak, dimensional, and penetrant inspection	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Dyno backpressure/noise/temperature correlation	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

SaintFlow Titanium Exhaust - Verification, Failure Modes, and Release

SF-19 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Pressure/leak and weld coupon validation	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Thermal cycle and shaker/vehicle durability	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Dyno torque/backpressure/EGT/noise	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Heat-shield temperature survey	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Track inspection for cracks, slip joints, and mounts	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Weld cracking	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Thermal bind	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Mount fatigue	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Packing/catalyst failure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Excessive heat transfer	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Noise/rule noncompliance	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Measured temperature-based material choice
- ☐ Free thermal growth without contact
- ☐ Replaceable modules
- ☐ Weld genealogy and NDT
- ☐ Event/market-specific compliant configuration

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

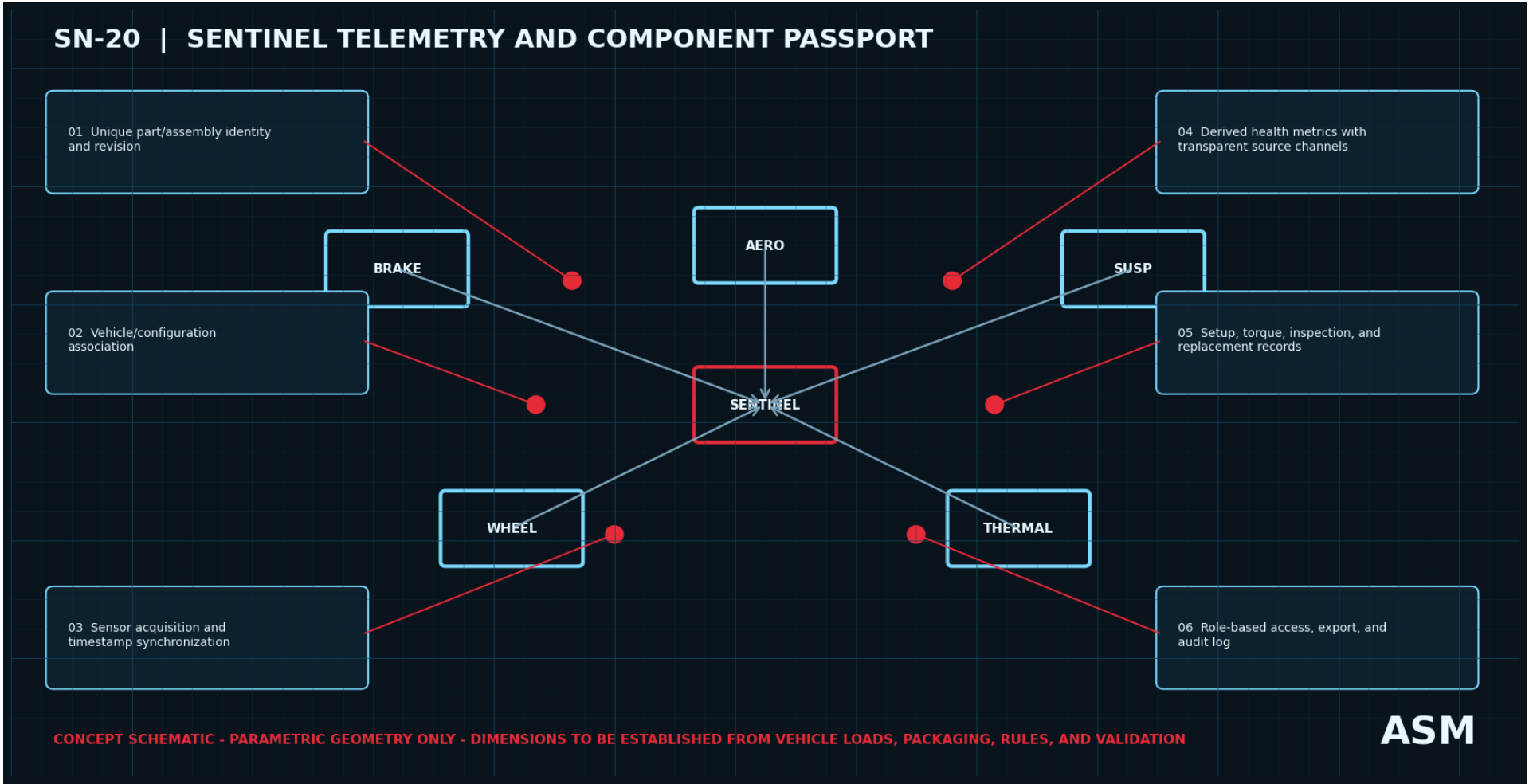
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

Sentinel Telemetry and Component Passport

SN-20 / SHEET 1 OF 4

SENTINEL SYSTEMS

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Connect ASM hardware, setup, condition, and life data into one traceable engineering system without allowing software to conceal unsafe mechanical states.	Component passports, Halo health score, configuration checksum, synchronized corner data, and service-gate alerts.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

Sentinel Telemetry and Component Passport - Architecture and Design Basis

SN-20 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Unique part/assembly identity and revision
- Vehicle/configuration association
- Sensor acquisition and timestamp synchronization
- Derived health metrics with transparent source channels
- Setup, torque, inspection, and replacement records
- Role-based access, export, and audit log

PRIMARY DESIGN VARIABLES

- Sampling rates, aliasing, synchronization, and sensor accuracy
- Data ownership and schema
- Plausibility, missing data, and calibration expiration
- Cybersecurity and access control
- Safe behavior when data or connectivity is unavailable
- Human-readable inspection workflow

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- All ASM systems
- ECU/data logger/CAN
- Workshop PLM/ERP/service tools
- Driver display
- Network/export interfaces

CALCULATION FRAMEWORK

- Sensor scaling and uncertainty budget
- State/threshold logic derived from verified limits
- Life usage accumulation from counted events/duty
- Checksums and immutable audit records

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

Sentinel Telemetry and Component Passport - Materials, Manufacturing, and Quality

SN-20 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Controller/logger	Motorsport-qualified hardware	Voltage, vibration, temperature, ingress
Sensors	Calibrated pressure/temp/strain/position units	Traceable calibration
Harness	Sealed automotive/motorsport wiring	Strain relief and EMI
Software	Version-controlled embedded/cloud/local stack	Change control and verification

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Define data dictionary and safety boundaries	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Bench hardware-in-loop simulator	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Calibration and channel verification tooling	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Threat analysis and secure update method	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Vehicle commissioning and controlled fault injection	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

Sentinel Telemetry and Component Passport - Verification, Failure Modes, and Release

SN-20 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Unit/integration/system software tests	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Hardware-in-loop and sensor fault injection	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Clock drift, dropped packets, corrupted data, and power cycling	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Cybersecurity assessment aligned to project scope	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Field usability and audit reconstruction	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
False safe indication	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Missed alarm	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Sensor drift	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Configuration mismatch	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Unauthorized change	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Data loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Mechanical inspection remains authoritative
- ☐ Every derived alert links to raw evidence
- ☐ Calibration expiration visible
- ☐ Signed configuration/revision
- ☐ Offline-safe operation and exportable audit trail

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

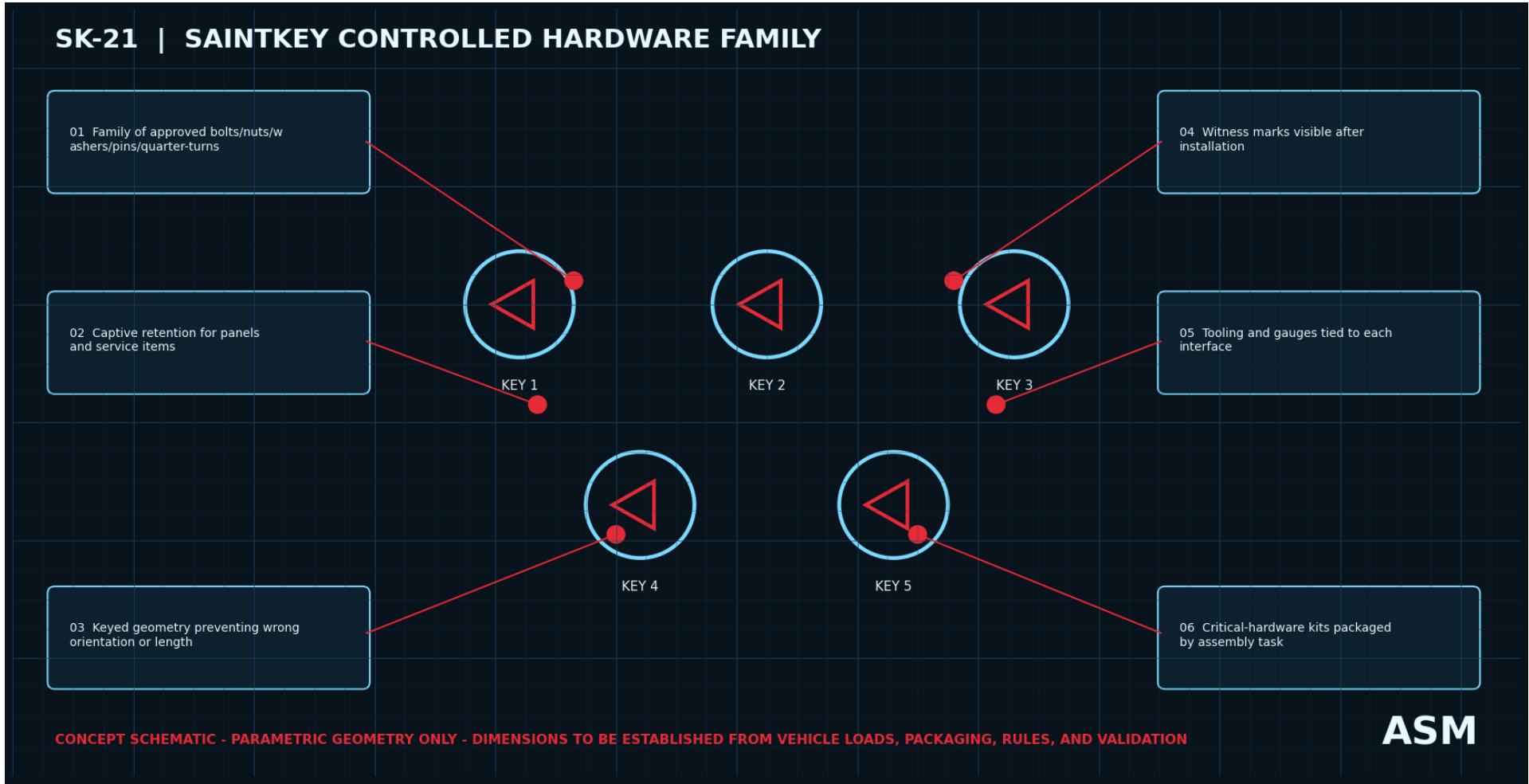
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

SaintKey Controlled Hardware Family

SK-21 / SHEET 1 OF 4

ASM PLATFORM

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Standardize fasteners, quick-release interfaces, witness marks, and service tools so ASM vehicles are faster to inspect and harder to assemble incorrectly.	Trinity tool interface, captive elements, red/white lock witnesses, keyed orientation, and serialized critical hardware kits.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

SaintKey Controlled Hardware Family - Architecture and Design Basis

SK-21 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Family of approved bolts/nuts/washers/pins/quarter-turns
- Captive retention for panels and service items
- Keyed geometry preventing wrong orientation or length
- Witness marks visible after installation
- Tooling and gauges tied to each interface
- Critical-hardware kits packaged by assembly task

PRIMARY DESIGN VARIABLES

- Clamp load and joint stiffness
- Grip length, thread engagement, and prevailing torque
- Temperature, corrosion, and galvanic environment
- Repeated service cycles and tool wear
- Foreign-object-debris risk
- Human factors under pit time pressure

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Every mechanical system
- Pit/workshop tooling
- PLM/BOM and Component Passport
- Supplier quality
- Scrutineering/service workflows

CALCULATION FRAMEWORK

- Joint preload and separation margin
- Bolt fatigue from external load fraction
- Bearing/pull-through at panels
- Torque scatter established experimentally

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

SaintKey Controlled Hardware Family - Materials, Manufacturing, and Quality

SK-21 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
Critical bolts	Certified alloy steel/titanium only where engineered	Full material/heat-treat traceability
Washers/inserts	Hardened alloy with surface treatment	Embedment and galling control
Captive panel hardware	Qualified motorsport supplier parts	Do not invent unsafe locks
Witness media	High-temperature tamper/witness coating	Compatible with substrate

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Create joint-class standard and approved-part database	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Torque-tension testing with real finishes/lubricants	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Gauge and tool design	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Assembly work instructions with photographs	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Reuse-cycle and inspection validation	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

SaintKey Controlled Hardware Family - Verification, Failure Modes, and Release

SK-21 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Proof/fatigue by joint class	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Vibration loosening and reuse-cycle tests	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	Misassembly trials with technicians	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Corrosion/heat/chemical compatibility	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Tool/socket wear and damaged-fastener removal	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Wrong fastener/length	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Preload loss	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Galling	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Captive feature failure	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Tool misengagement	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Unrecorded reuse	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ No cosmetic-only proprietary critical thread
- ☐ Approved-source list
- ☐ Visual ID without relying on color alone
- ☐ Defined reuse/discard rules
- ☐ Task kits and digital work instruction

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

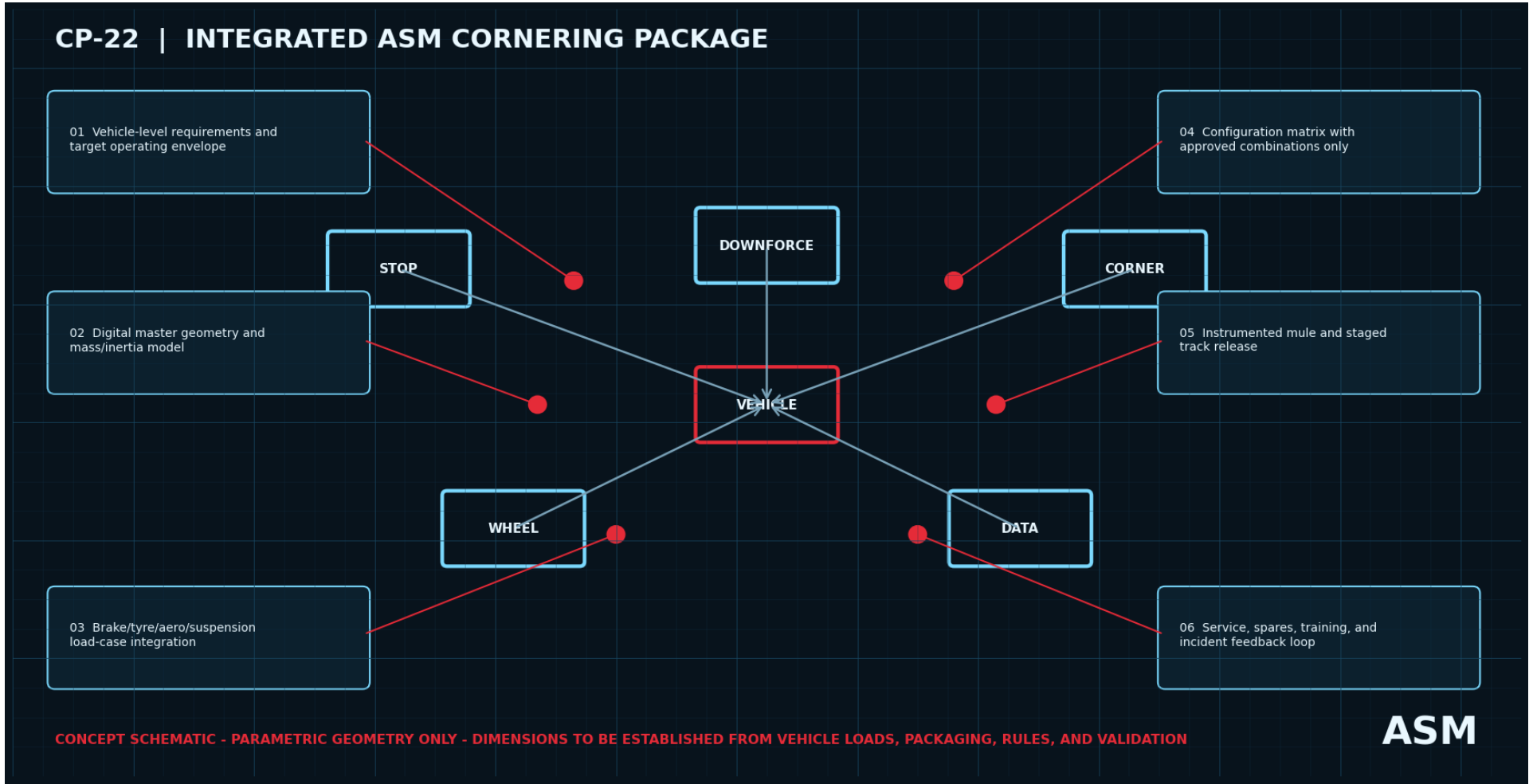
Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

Integrated ASM Cornering Package

CP-22 / SHEET 1 OF 4

ASM PLATFORM

Concept schematic and design identity. The figure communicates architecture and interfaces; it is not a manufacturing drawing and intentionally omits unvalidated dimensions.



Design intent	ASM-specific execution	Release principle
Release braking, wheel, suspension, and aero changes as a measured vehicle system rather than independent catalog parts.	One setup passport linking HaloBridge, HaloFlow, HaloLock, HaloRail, VectorBar, CornerBlade, SaintLink, VortexRail, and HaloArc.	No production release until requirements, drawings, calculations, supplier process, DV/PV evidence, and independent approval are linked in the configuration record.

Integrated ASM Cornering Package - Architecture and Design Basis

CP-22 / SHEET 2 OF 4

SYSTEM ARCHITECTURE

- Vehicle-level requirements and target operating envelope
- Digital master geometry and mass/inertia model
- Brake/tyre/aero/suspension load-case integration
- Configuration matrix with approved combinations only
- Instrumented mule and staged track release
- Service, spares, training, and incident feedback loop

PRIMARY DESIGN VARIABLES

- Lap-time objective balanced against stability and durability
- Tyre load/temperature and brake energy
- Aero balance versus speed/ride/yaw
- Roll/pitch/heave platform and driver confidence
- Unsprung/rotating mass and compliance
- Failure interactions and degraded states

PARAMETERIZATION RULE

Use named parameters rather than fixed copied dimensions. Establish a vehicle coordinate system and master interface drawing. The CAD model must derive critical geometry from controlled parameters, include tolerance envelopes, and produce an interface-control drawing that can be reviewed independently of the native CAD file.

Gate	Required evidence
A - Feasibility	Packaging study; first-principles load/thermal/flow estimate; prior-art and rules review; hazards list.
B - Concept freeze	Trade study; selected architecture; preliminary FEA/CFD/1D model; interface-control drawing; prototype plan.
C - Detail design	Released drawings; tolerance analysis; material/process specifications; supplier plan; DFMEA; verification procedures.

INTERFACE CONTROL

- Driver and controls
- Tyres and vehicle platform
- All ASM subsystems
- Rules/homologation
- Manufacturing and service organization

CALCULATION FRAMEWORK

- Brake energy and balance model
- Quasi-static load transfer and tyre load sensitivity
- Aero force/balance model
- Ride/roll stiffness and frequency model
- Lap simulation used as ranking tool, not validation substitute

REQUIREMENTS TO FREEZE BEFORE DETAIL DESIGN

- Target vehicle mass, axle loads, tire capability, speed, duty cycle, ruleset, and service life.
- Nominal and worst-case hard points, envelopes, temperatures, pressures, manufacturing variation, and misuse loads.
- Safety goals, allowable failure states, inspection interval, and replacement criteria.
- Quantitative acceptance values; every value must have an owner, source, method, and verification activity.

Integrated ASM Cornering Package - Materials, Manufacturing, and Quality

CP-22 / SHEET 3 OF 4

MATERIAL CANDIDATES - FINAL SELECTION REQUIRES COUPON AND COMPONENT DATA

Element	Candidate	Engineering reason / control
System BOM	Approved materials from each subsystem	No substitutions without change review
Mule hardware	Production-representative where loads matter	Instrument heavily
Instrumentation	Pressure/temp/strain/load/position	Uncertainty documented
Documentation	Controlled drawings, code, setup and test records	Single source of truth

PROTOTYPE-TO-PRODUCTION PROCESS

MINIMUM QUALITY PLAN

Stage	Development action	Deliverable	Characteristic	Control method	Typical record
1	Freeze baseline vehicle and collect repeatable data	Material/process se record	Material identity	Certificate review plus positive material identification where appropriate	Heat/lot certificate linked to serial number
2	Simulate and bench-test subsystem candidates	Controlled CAD/dra process spec	Datums and geometry	CMM, calibrated gauges, scan, runout, or fixture appropriate to feature	Ballooned inspection report
3	Install one controlled change at a time initially	Inspection or test evidence	Surface and edge condition	Roughness measurement, visual standard, deburr/edge-break inspection	Process traveler and photos
4	Progress from proving ground to closed track to endurance	Configuration and genealogy record	Internal integrity	Penetrant, magnetic particle, ultrasound, CT, tap test, or proof test as applicable	NDT report with acceptance standard
5	Release only approved combination with service package	Nonconformance/re disposition	Assembly state	Torque/preload/adhesive cure/float/calibration check	Serialized assembly traveler

DRAWING PACKAGE MINIMUM

Assembly drawing; detail drawings; interface-control drawing; material and finish specifications; general tolerances and geometric tolerancing per the selected company standard; critical-characteristic symbols; inspection plan; approved supplier/process list; assembly traveler; service instructions; life/inspection labels; and a revision-controlled bill of materials.

Integrated ASM Cornering Package - Verification, Failure Modes, and Release

CP-22 / SHEET 4 OF 4

DESIGN VERIFICATION MATRIX

Level	Verification activity	Pass evidence
Analysis	Requirements traceability review	Preapproved procedure, calibrated data, anomalies disposition, signed report
Bench	Subsystem DV plus system integration test	Preapproved procedure, calibrated data, anomalies disposition, signed report
Component DV	K&C;, corner-weight, brake dyno, aero correlation, and track test	Preapproved procedure, calibrated data, anomalies disposition, signed report
Vehicle DV	Endurance/abuse and post-test teardown	Preapproved procedure, calibrated data, anomalies disposition, signed report
Release/PV	Independent design/safety review before customer release	Preapproved procedure, calibrated data, anomalies disposition, signed report

PRELIMINARY FAILURE-MODE REGISTER

Potential failure	Prevention/detection concept	Release requirement
Beneficial subsystem interaction assumed but absent	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Unstable balance migration	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Thermal overload cascade	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Configuration mismatch	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Maintenance complexity	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted
Driver overload or poor feedback	Design margin + process control + inspection/test + clear service trigger	Severity/occurrence/detection reviewed; action closed or formally accepted

CONCEPT ACCEPTANCE TARGETS

- ☐ Approved combination matrix
- ☐ Baseline-versus-ASM evidence
- ☐ No release from simulation alone
- ☐ Independent review and signed test report
- ☐ Incident reporting and continuous improvement

RELEASE HOLD POINTS

- Independent calculation/model review.
- Drawing and tolerance review.
- Supplier process-readiness review.
- Design validation report approved.
- Production validation on representative parts.
- Vehicle integration and rules/compliance review.
- Service, inspection, retirement, and incident procedures published.

SAFETY STATEMENT

Any brake, wheel-retention, suspension, steering-adjacent, structural, fuel/air, exhaust, or high-temperature component is safety critical. Prototype use must be limited to controlled facilities under a written test plan, with inspection before and after each run.

Appendix A - System Requirements Specification Template

APPENDIX A

Field	Required content
Identity	System code, product name, revision, owner, vehicle application, intended rules/market.
Use cases	Road/track/race duty, speed, mass, tire, environment, driver, session/endurance, service assumptions.
Functional requirements	Numbered "shall" statements with metric, condition, tolerance, and verification method.
Safety requirements	Hazards, severity, safe/degraded state, detection, inspection, and fault response.
Interfaces	Datums, geometry, loads, temperatures, pressures, fluids, electrical/network, software/calibration.
Life and maintenance	Duty-cycle definition, inspection method/interval, wear limits, life limit, storage and transport.
Manufacturing	Material, process, special processes, critical characteristics, capability and genealogy.
Compliance	Rules/standards applicability matrix and approval owner.
Verification	Analysis/inspection/test/demonstration with pass values and report IDs.
Change control	Change classes, re-analysis/retest rules, approved substitutions, and field-update path.

EXAMPLE REQUIREMENT FORM

ID	Requirement	Condition	Acceptance	Method	Evidence
SYS-###	The assembly shall ...	At specified load, temperature, tolerance, and life state	Numeric limit with guard band	Analysis / inspection / test	Report or record ID

Appendix B - DFMEA and Hazard Review Template

APPENDIX B

Item/function	Failure mode	Effect	Cause/mechanism	Prevention control	Detection control	Action/owner	Evidence/closure
Example: joint maintains preload	Preload lost	Movement, fatigue, detachment	Friction scatter, embedment, temperature, misuse	Joint model, surface/lube spec, installation method	Witness, torque audit, inspection, test	Redesign/test/process action	Closed report and revised drawing

HAZARD QUESTIONS

What can detach, lock, leak, ignite, puncture, overheat, collapse, mislead, or change vehicle balance? What happens after one fastener, sensor, seal, hose, bearing, or communication path fails?

DETECTION QUESTIONS

Can the crew see, feel, measure, or inspect the degradation before it becomes dangerous? Is the indication independent of the function being monitored?

SERVICE QUESTIONS

Can a rushed technician install it backwards, omit a spacer, reuse a life-limited item, apply the wrong lubricant, or select the wrong calibration? Design the error out where practical.

SCORING NOTE

Appendix C - Design Verification Plan and Report Template

APPENDIX C

Requirement ID	Test article/configuration	Method/procedure	Instrumentation and uncertainty	Loads/environment/duration	Pass criteria	Result/anomalies	Report/approval
SYS-###	Serial numbers; drawing/BOM/software revisions	Approved procedure ID	Sensor IDs, calibration dates, uncertainty	Exact duty and sequence	Predefined numeric limit	Pass/fail plus anomaly disposition	Signed report and raw-data location

TEST ARTICLE CONTROL

Photograph and identify every article. Record material lots, supplier processes, deviations, assembly values, sensor calibration, fluids, software, setup, pre-test inspection, test interruptions, post-test condition, teardown, NDT, and retained samples. Never discard a failed part until the failure review is complete.

ANOMALY RULE

A test is not automatically valid because the measured result passed. Review fixture failures, sensor saturation, unexpected noise, control excursions, interruptions, data gaps, and nonrepresentative article conditions. Anomalies require documented disposition and may require rerun.

CORRELATION RULE

Compare predicted and measured loads, strain, displacement, pressure, temperature, flow, or frequency at defined locations. Record model version, boundary conditions, material data source, and calibration changes. Use correlation to improve future predictions, not to retroactively hide an incorrect requirement.

Appendix D - Drawing, Tolerance, and Release Checklist

APPENDIX D

Area	Release checklist
Identification	Part/assembly number, description, revision, units, scale, sheet, owner, approvals, application.
Datums	Functional datum reference frame tied to interfaces and inspection access.
Geometry	Critical fits, runout, profile, position, flatness, parallelism, concentricity only where function requires.
Materials/processes	Exact specification, temper/condition, heat treatment, coating, masking, welding/bonding, cure, shot peen, NDT.
Edges/surfaces	Deburr/edge break, no sharp edges, surface roughness, sealing and friction surfaces, cosmetic limits separately identified.
Threads/joints	Standard thread callout, engagement, insert, locking method, lubricant, torque/preload process located in controlled work instruction.
Composites	Ply book, material batch, orientation, overlaps, core/closeouts, inserts, adhesive, cure, trim, NDT and repair limits.
Inspection	Critical characteristic symbols, gauges/fixtures, CMM strategy, sampling, capability, first-article requirements.
Service	Orientation, assembly sequence, special tools, witness marks, inspection, wear/discard, life, storage, and replacement.
Digital continuity	CAD/drawing/BOM/software/test evidence all refer to the same released configuration.

TOLERANCE PRINCIPLE

Appendix E - Manufacturing Launch and Supplier Quality

APPENDIX E

Phase	Required outputs
Supplier selection	Capability audit; special-process approvals; equipment; metrology; capacity; traceability; change-notification terms.
Process development	Flow chart; PFMEA; tooling/fixtures; work instructions; inspection plan; packaging and preservation.
First article	Ballooned drawing report; certificates; process records; NDT; functional/proof test; deviations closed.
Capability	Critical features measured over representative production; stability and capability reviewed before volume release.
Production validation	Production tooling, operators, materials, inspection, software, and rate used to build/test representative lots.
Ongoing control	Lot/serial genealogy; calibration; preventive maintenance; control charts where useful; nonconformance and escape response.
Change management	No supplier, material, tooling, heat-treat, finish, program, inspection, or sub-tier change without prior approval.
Field feedback	Returned-part quarantine, teardown, root cause, containment, corrective action, and life-limit updates.

SPECIAL PROCESSES

Heat treatment, welding, brazing, adhesive bonding, composite cure, plating/coating, shot peening, and NDT cannot be verified completely by final dimensional inspection. They require qualified procedures, trained personnel, controlled equipment, coupons or witness specimens where appropriate, and retained process records.

Appendix F - Intellectual Property and ASM Exclusivity

APPENDIX F

Protection layer	Candidate ASM subject matter	Action
Utility patents	Functional bridge preload, flow cartridges, locking/inspection mechanisms, modular stiffness or aero systems, sensor/load integration.	Prior-art search; document inventors/dates; file before public disclosure where strategy supports it.
Design patents/registered designs	HaloBridge form, Trinity nut, HaloRail wheel/spokes, HaloArc planform/mounts, Aero Vein patterns.	Create consistent render/drawing set and jurisdiction plan.
Trademarks	ASM division and product-family names, symbols, service marks.	Clearance search and registration strategy.
Trade secrets	Material tuning, friction formulations, CFD correlations, manufacturing parameters, setup maps, algorithms.	Need-to-know access, NDAs, supplier clauses, security, employee invention assignment.
Copyright/database rights	Manuals, CAD drawings, software, data schema, setup/test database.	Ownership and license clauses; controlled distribution.
Trade dress	Consistent Halo/Trinity/Aero Vein visual language across products.	Use consistently and avoid purely ornamental claims for functional shapes.

OWNERSHIP RULE

Every employee, contractor, university partner, supplier, and test laboratory agreement should state ownership of inventions, CAD, tooling, software, raw data, reports, improvements, background IP, and rights to reuse. Commercial confidentiality must not prevent ASM from accessing safety and quality evidence.

Appendix G - Proposed 36-Month Development Sequence

APPENDIX G

Period	Focus	Systems and milestones
0-3 months	Foundation	Vehicle/use-case selection; baseline instrumentation; requirements system; product names/IP screen; supplier/lab RFQs.
3-9 months	First corner mule	HaloFlow prototypes; pad/rotor screening; HaloBridge architecture; control-arm/upright scans and load acquisition; Sentinel data dictionary.
9-15 months	Bench evidence	Caliper/rotor/pad dyno; center-lock joint rig; wheel FEA/supplier mule; sway-bar calibration; wing/splitter CFD and proof fixtures.
15-21 months	Integrated mule	Approved prototype corner package; K&C;/corner rig; aero load measurement; thermal network; controlled low/high-speed track progression.
21-27 months	Endurance and production intent	Representative processes; DV endurance; teardown/NDT; design updates; supplier control plans; service documentation.
27-33 months	PV and compliance	First articles and capability; production validation; rules/homologation/legal evidence for selected market/application.
33-36 months	Limited release	Trained installers; serialized kits; monitored customer/works-car program; strict incident and inspection feedback.

SEQUENCING PRINCIPLE

Develop the first product around one defined vehicle and one use case. Universal fitment multiplies interfaces, loads, legal classifications, and test combinations. A narrower application with excellent evidence is a stronger ASM launch than a broad catalog of unvalidated concepts.

Appendix H - Glossary and Reference Register

APPENDIX H

GLOSSARY

Term	Meaning
DFMEA / PFMEA	Design / Process Failure Modes and Effects Analysis.
DV / PV	Design validation / production validation.
FAI	First Article Inspection.
FTO	Freedom-to-operate intellectual-property review.
ICD	Interface Control Document or Drawing.
K&C;	Kinematics and compliance testing.
NDT	Nondestructive testing.
PLM	Product Lifecycle Management system.
Proof / ultimate	Project-defined non-destructive proof and failure/ultimate load concepts; exact factors require applicable discipline and rules.
Traceability	Ability to link part identity to revision, materials, processes, inspection, assembly, test, service, and changes.

REFERENCE REGISTER

Reference register - verify current edition before use
FIA 2026 Regulations portal and Appendix J, Article 253 - safety equipment and technical requirements.
NHTSA Laboratory Test Procedure for FMVSS No. 135 - Light Vehicle Brake Systems.
SAE J2522_202312 - Dynamometer Global Brake Effectiveness.
SAE J2530_202412 - Aftermarket Wheels - Passenger Cars and Light Truck - Performance Requirements and Test Procedures.
UNECE Regulation No. 90 - replacement braking components.
ISO 26262:2018 - Road vehicles - Functional safety.
ISO/SAE 21434:2021 - Road vehicles - Cybersecurity engineering.
ASME Y14.5 - Dimensioning and Tolerancing.
Vehicle, sanctioning-body, supplier, and material/process standards applicable to the selected project.

END OF CONTROLLED CONCEPT MANUAL